Biomathematics 2

– Lecture notes –

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Preface

These notes are intended for graduate students in applied mathematics and cover basic topics of mathematical modeling of biological systems, with a special emphasis on the description of the membrane of excitable cells and of excitable tissues, such as the cardiac muscle.

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Chapter 1
Biochemical reactions

1.1 Law of Mass Action

The fundamental ”law” of a chemical reaction is the law of mass action. This law describes the rate at which chemicals, whether large macromolecules or simple ions, collide and interact to form different chemical combinations. Suppose that two chemicals, say A and B, react upon collision with each other to form product C.

Denoting by $a(t),\ b(t),\ c(t)$ the concentrations of A, B, C, as function of time, respectively, the rate of this reaction is the rate of accumulation of product, i.e.

$$\frac{dc(t)}{dt}.$$

This rate is the product of the number of collisions per unit time between the two reactants and the probability that a collision is sufficiently energetic to overcome the free energy of activation of the reaction. The number of collisions per unit time is taken to be proportional to the product of the concentrations of A and B with a factor of proportionality that depends on the geometrical shapes and sizes of the reactant molecules and on the temperature of the mixture.

In formulae, the *law of mass action* reads:

$$\frac{dc(t)}{dt} = k_+ a(t)b(t)$$

with $k_+ > 0$ constant.

We call *diagram of the reaction* the scheme:

$$\mathrm{A} + \mathrm{B} \xrightarrow{k_+} \mathrm{C}$$

The global dynamisc of the reaction is modeled by the system of ordinary differential equations (ODEs):

$$\begin{cases} \dfrac{da(t)}{dt} = -k_+ a(t)b(t) \\ \dfrac{db(t)}{dt} = -k_+ a(t)b(t) \\ \dfrac{dc(t)}{dt} = k_+ a(t)b(t) \end{cases}$$

with initial conditions $a(0) = a_0 > 0$, $b(0) = b_0 > 0$, $c(0) = c_0 = 0$.

Consider now the following reversible reaction:

$$\mathrm{A} + \mathrm{B} \underset{k_-}{\overset{k_+}{\rightleftarrows}} \mathrm{C}$$

The associated system of ODEs reads:

$$\begin{cases} \dfrac{da(t)}{dt} = -k_+ a(t)b(t) + k_- c(t) \\ \dfrac{db(t)}{dt} = -k_+ a(t)b(t) + k_- c(t) \\ \dfrac{dc(t)}{dt} = k_+ a(t)b(t) - k_- c(t) \end{cases}$$

with initial conditions $a(0) = a_0 > 0$, $b(0) = b_0 > 0$, $c(0) = c_0 = 0$ and $k_+ > k_- > 0$.

Observe that

$$\frac{d(a(t)+c(t))}{dt} = \frac{d(b(t)+c(t))}{dt} = 0,$$

hence the quantities

$$a(t) + c(t) \qquad \text{and} \qquad b(t) + c(t)$$

conserve themselves. Therefore

$$\begin{cases} a(t) + c(t) = a(0) + c(0) = a_0 \\ b(t) + c(t) = b(0) + c(0) = b_0. \end{cases}$$

By setting

$$\begin{cases} a(t) = a_0 - c(t) \\ b(t) = b_0 - c(t), \end{cases}$$

we eliminate the equations for $a(t)$ and $b(t)$, obtaining only the following Riccati equation for $c(t)$:

$$\begin{aligned} \frac{dc(t)}{dt} &= k_+(a_0 - c(t))(b_0 - c(t)) - k_- c(t) \\ &= k_+ c^2(t) - (k_+ a_0 + k_+ b_0 + k_-)c(t) + k_+ a_0 b_0 \end{aligned}$$

with initial condition $c(0) = c_0 = 0$.

1.2 Enzyme kinetics

In order to see where some of the more complicated reaction schemes come from, we consider a reaction that is catalyzed by an enzyme. Enzymes are catalysts (generally proteins) that help convert other molecules called substrates into products, but they themselves are not changed by the reaction. Their most important features are catalytic power, specificity and regulation. Enzymes accelerate the conversion of substrate into product by lowering the free energy of activation of the reaction. For example, enzymes may aid in overcoming charge repulsions and allowing reacting molecules to come into contact for the formation of new chemical bonds. Or, if the reaction requires breaking of an existing bond, the enzyme may exert a stress on a substrate molecule, rendering a particular bond more easily broken. Enzymes are particularly efficient at speeding up biological reactions, giving increases in speed of up to 10 million times or more. They are also highly specific, usually catalyzing the reaction of only one particular substrate or closely related substrates. Finally, they are typically regulated by an enormously complicated set of positive and negative feedbacks, thus allowing precise control over the rate of reaction.

The first simple model for enzyme reactions was proposed by Michaelis and Menten (1913). In their reaction scheme, the enzyme E converts the substrate S into the product P through a two-step process. First E combines with S to form a complex C which then breaks down into the product P releasing E in the process. The reaction scheme is represented schematically by

$$\mathrm{S} + \mathrm{E} \underset{k_{-1}}{\overset{k_1}{\rightleftarrows}} \mathrm{C} \overset{k_2}{\rightarrow} \mathrm{P} + \mathrm{E}$$

Let $s,\ c,\ e,\ p$ be the concentrations of S, C, E, P, respectively. By applying the law of mass action, we obtain the ODE system:

$$\begin{cases} \dfrac{ds}{dt} = k_{-1}c - k_1 es & s(0) = s_0 \\ \dfrac{de}{dt} = (k_{-1} + k_2)c - k_1 es & e(0) = e_0 \\ \dfrac{dc}{dt} = k_1 es - (k_{-1} + k_2)c & c(0) = 0 \\ \dfrac{dp}{dt} = k_2 c & p(0) = 0 \end{cases} \tag{1.1}$$

The following *conservation properties* hold:

- ***Enzyme conservation***:

$$\frac{de}{dt} + \frac{dc}{dt} = 0 \Rightarrow e = e_0 - c;$$

- ***Substrate conservation***:

$$\frac{ds}{dt} + \frac{dc}{dt} + \frac{dp}{dt} = 0 \Rightarrow s + c + p = s_0.$$

The quantity

$$v(t) = \frac{dp}{dt} = k_2 c$$

is called ***reaction velocity***.

By substituting in (1.1) the expression for e and eliminating the last equation we obtain the system:

$$\begin{cases} \dfrac{ds}{dt} = -k_1 e_0 s + (k_1 s + k_{-1})c \qquad s(0) = s_0 \\ \dfrac{dc}{dt} = k_1 e_0 s - (k_{-1} + k_2 + k_1 s)c \;\; c(0) = 0 \end{cases} \tag{1.2}$$

The dynamics of the previous system is characterized by two states: the *transient state* (regime transitorio) and the *quasi-steady state* (regime permanente), see Fig. 1.1.

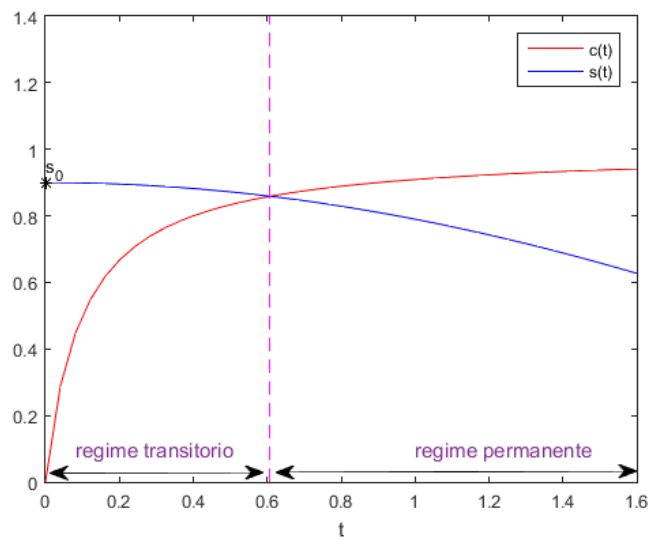


Fig. 1.1: Enzyme kinetics: *transient state* (regime transitorio) and *quasi-steady state* (regime permanente).

1.2.1 Equilibrium approximation

We assume that the substrate is in instantaneous equilibrium with the complex, and thus

$$k_1 se = k_{-1}c \quad \Rightarrow \frac{ds}{dt} = 0.$$

With this assumption, it follows that $s = s_0$ and from the first equation in (1.2):

$$-k_1 e_0 s_0 + (k_1 s_0 + k_{-1})c = 0.$$

Therefore we obtain

$$c = \frac{k_1 e_0 s_0}{k_1 s_0 + k_{-1}} = \frac{k_1 e_0 s_0}{k_1\left(s_0 + \frac{k_{-1}}{k_1}\right)} = \frac{e_0 s_0}{s_0 + k_M},$$

with $k_M := \frac{k_{-1}}{k_1}$. The *reaction velocity* is:

$$v = \frac{dp}{dt} = k_2 c = k_2 \frac{e_0 s_0}{s_0 + k_M} = \frac{v_{max} s_0}{k_M + s_0},$$

where $v_{max} = k_2 e_0$ is the maximum reaction velocity, which is obtained when all the enzyme is bound to the substrate. The expression

$$v = \frac{v_{max} s_0}{k_M + s_0}$$

is known as ***Michaelis-Menten law***.

1.2.2 Quasi-steady state approximation

This approximation holds in the *quasi-steady state*, where we assume that the rates of formation and breakdown of the complex are essentially equal at all times. Thus

$$\frac{dc}{dt} = 0.$$

System (1.2) becomes

$$\begin{cases} \dfrac{ds}{dt} = -k_1 e_0 s + (k_1 s + k_{-1})c \quad s(0) = s_0 \\ k_1 e_0 s - (k_{-1} + k_2 + k_1 s)c = 0, \end{cases} \tag{1.3}$$

from which it follows

$$c = \frac{k_1 e_0 s}{k_{-1} + k_2 + k_1 s} = \frac{k_1 e_0 s}{k_1 \left(s + \frac{k_{-1}+k_2}{k_1}\right)} = \frac{e_0 s}{s + k_m},$$

with $k_m := \frac{k_{-1}+k_2}{k_1}$. Moreover, from the substrate conservation property we have:

$$\frac{ds}{dt} = -\frac{dc}{dt} - \frac{dp}{dt} = -\frac{dp}{dt} = -k_2 c = -k_2 \frac{e_0 s}{s + k_m} \quad \Rightarrow \frac{ds}{dt} = -\frac{v_{max} s}{s + k_m}.$$

By integrating the previous equation, we obtain the implicit formula:

$$s(t) + k_m log(s(t)) = s_0 + k_m log(s_0) - v_{max} t.$$

The reaction velocity is given by

$$v = \frac{dp}{dt} = k_2 c = \frac{v_{max} s}{s + k_m}.$$

The Michaelis-Menten equilibrium constant $k_m := \frac{k_{-1}+k_2}{k_1}$ is important because of

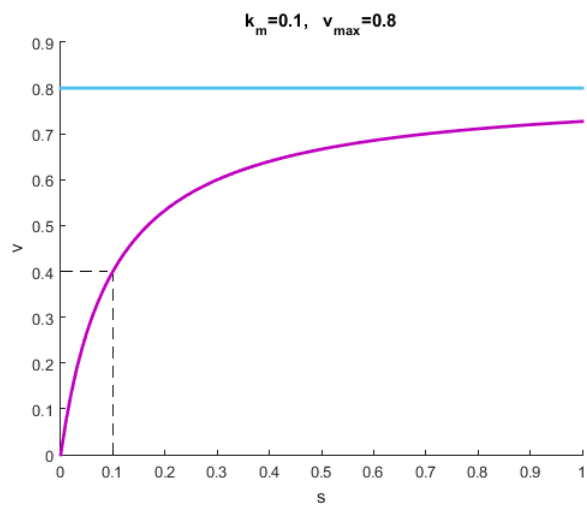


Fig. 1.2: Michaelis-Menten law.

the following reasons:

- it is a characteristic quantity of each enzyme;
- it represents the substrate concentration necessary for the reaction to have a velocity equal to half of the maximum velocity: indeed by imposing $v = \frac{1}{2} v_{max}$, we have:

$$\frac{v_{max}}{2} = \frac{v_{max}s}{s+k_m} \quad \Rightarrow \quad k_m + s = 2s \quad \Rightarrow \quad s = k_m;$$

- it measures the affinity of the enzyme for the substrate:

$$\text{low } k_m \Rightarrow \text{high affinity}$$
$$\text{high } k_m \Rightarrow \text{low affinity}$$

From the Michaelis-Menten law, giving the reaction velocity as a function of the substrate concentration (see Fig. 1.2), we derive the following considerations:

1. At low substrate concentrations, the reaction velocity increases proportionally with s;
2. At high substrate concentrations, the reaction velocity reaches a constant maximum value. This is due to the complete saturation of the enzyme which cancels the effect due to the further increase in the substrate concentration. In this case $v_{max} = k_2 e_0$.

1.2.3 Time constants

During the *transient state*, dynamics is mainly determined by equation

$$\frac{dc}{dt} = -(k_{-1} + k_2 + k_1 s_0)c + k_1 e_0 s_0.$$

Since $k_{-1} + k_2 << k_1 s_0$, it follows $k_{-1} + k_2 + k_1 s_0 \approx k_1 s_0$. Therefore the time constant of the *transient state* is given by

$$\tau_r = \frac{1}{k_1 s_0}.$$

On the other hand, during the *quasi-steady state*, dynamics is mainly determined by equation

$$\frac{ds}{dt} = -k_1 e_0 s + (k_1 s + k_{-1})c.$$

Therefore the time constant of the *quasi-steady state* is given by

$$\tau_p = \frac{1}{k_1 e_0}.$$

We remark that the *transient state* is much faster than the *quasi-steady state*, because

$$e_0 << s_0 \qquad \Rightarrow \tau_r << \tau_p.$$

1.2.4 Uniform approximation

Scope of this section is to construct an approximation of system (1.2) holding not only during the *quasi-steady state* but also during the *transient state*. We call such approximation ***Uniform Approximation***, whose construction is based of the following steps:

1. Derivation of the dimensionless system
2. Construction of outer solution
3. Construction of inner solution
4. Matching

1.2.4.1 Adimensionalized system

Let $\varepsilon := \frac{e_0}{s_0}$ and $\tau = \frac{t}{\tau_p} = k_1 e_0 t$, which implies $\frac{dt}{d\tau} = \frac{1}{k_1 e_0}$ and consider

$$\textit{dimensionless s:} \quad u(\tau) = \frac{s(t)}{s_0} \quad \Rightarrow s(t) = s_0 u(\tau)$$

$$\textit{dimensionless c:} \quad v(\tau) = \frac{c(t)}{e_0} \quad \Rightarrow c(t) = e_0 v(\tau).$$

By deriving u and v, we obtain:

$$\frac{du}{d\tau} = \frac{1}{s_0}\frac{ds}{dt}\frac{dt}{d\tau} = \frac{1}{k_1 e_0 s_0}\frac{ds}{dt} \quad \Rightarrow \frac{ds}{dt} = k_1 e_0 s_0 \frac{du}{d\tau}$$

$$\frac{dv}{d\tau} = \frac{1}{e_0}\frac{dc}{dt}\frac{dt}{d\tau} = \frac{1}{k_1 e_0^2}\frac{dc}{dt} \quad \Rightarrow \frac{dc}{dt} = k_1 e_0^2 \frac{dv}{d\tau}$$

By substituting in system (1.2) the expressions for $\frac{ds}{dt}$ and $\frac{dc}{dt}$, we get:

$$\begin{cases} k_1 e_0 s_0 \dfrac{du}{d\tau} = -k_1 e_0 s_0 u(\tau) + \big(k_1 s_0 u(\tau) + k_{-1}\big) e_0 v(\tau) \\ k_1 e_0^2 \dfrac{dv}{d\tau} = -\big(k_{-1} + k_2 + k_1 s_0 u(\tau)\big) e_0 v(\tau) + k_1 e_0 s_0 u(\tau) \end{cases}$$

Then dividing both equations by $k_1 e_0 s_0$, we obtain:

$$\begin{cases} \dfrac{du}{d\tau} = -u(\tau) + \left(u(\tau) + \dfrac{k_{-1}}{k_1 s_0}\right) v(\tau) & u(0) = \dfrac{s(0)}{s_0} = 1 \\ \dfrac{e_0}{s_0}\dfrac{dv}{d\tau} = -\left(\dfrac{k_{-1}+k_2}{k_1 s_0} + u(\tau)\right) v(\tau) + u(\tau) & v(0) = \dfrac{c(0)}{e_0} = 0 \end{cases}$$

Since $\varepsilon := \frac{e_0}{s_0}$ and denoting by $\alpha := \frac{k_{-1}}{k_1 s_0}$, $K := \frac{k_{-1}+k_2}{k_1 s_0}$, the ***dimensionless system*** reads:

$$\begin{cases} \dfrac{du}{d\tau} = -u + (u+\alpha)v & u(0) = 1 \\ \varepsilon \dfrac{dv}{d\tau} = -(K+u)v + u & v(0) = 0 \end{cases} \tag{1.4}$$

This system is a *singular perturbation problem*, since letting $\varepsilon = 0$, the system becomes a differential-algebraic equations system instead of an ordinary differential equations system.

Finding a uniform approximation for (u,v) means finding $(\tilde{u},\tilde{v})$ such that

$$\begin{cases} |u(\tau) - \tilde{u}(\tau)| = O(\varepsilon) \\ |v(\tau) - \tilde{v}(\tau)| = O(\varepsilon) \end{cases} \qquad \forall \tau \tag{1.5}$$

1.2.4.2 Outer solution

Let us seek for solutions u and v of the form:

$$u(\tau) = \sum_{n=0}^{+\infty} \varepsilon^n u_n(\tau)$$

$$v(\tau) = \sum_{n=0}^{+\infty} \varepsilon^n v_n(\tau)$$

Substituting such expressions in system (1.4), we obtain for $n = 0$:

$$\begin{cases} \dfrac{du_0}{d\tau} = -u_0 + (u_0+\alpha)v_0 & u_0(0) = 1 \\ 0 = -(K+u_0)v_0 + u_0 \end{cases}$$

We observe that v_0 <u>does not</u> satisfy the initial condition because

$$v_0(0) = \frac{u_0(0)}{K+u_0(0)} = \frac{1}{1+K} \neq 0.$$

Since v_0 does not satisfy the initial condition, (1.5) does not hold, hence (u_0,v_0) is not the uniform solution.

However, (u_0,v_0) is the *dimensionless quasi-steady state approximation*, i.e.

$$\begin{cases} |u(\tau) - u_0(\tau)| = O(\varepsilon) \\ |v(\tau) - v_0(\tau)| = O(\varepsilon) \end{cases} \qquad \underline{\forall \tau > \tau_0 > 0}$$

(u_0,v_0) is called *outer solution*, because it is a good approximation of (u,v) only outside the transient state.

1.2.4.3 Inner solution

We construct now an approximation of (u,v) that holds inside the transient state. To do so, we perform a change of variables by introducing the quantity $\sigma = \frac{\tau}{\varepsilon} = k_1 s_0 t$, hence

$$U(\sigma) = u(\tau) \quad \Rightarrow \frac{dU}{d\sigma} = \frac{du}{d\tau}\frac{d\tau}{d\sigma} = \varepsilon\frac{du}{d\tau} \quad \Rightarrow \frac{du}{d\tau} = \frac{1}{\varepsilon}\frac{dU}{d\sigma}$$
$$V(\sigma) = v(\tau) \quad \Rightarrow \frac{dV}{d\sigma} = \frac{dv}{d\tau}\frac{d\tau}{d\sigma} = \varepsilon\frac{dv}{d\tau} \quad \Rightarrow \frac{dv}{d\tau} = \frac{1}{\varepsilon}\frac{dV}{d\sigma}$$

Substituting the expressions of $\frac{du}{d\tau}$ and $\frac{dv}{d\tau}$ in system (1.4) and multiplying times ε the first equation, we obtain:

$$\begin{cases} \dfrac{dU}{d\sigma} = \varepsilon(-U + (U+\alpha)V) & U(0) = 1 \\ \dfrac{dV}{d\sigma} = -(K+U)V + U & V(0) = 0 \end{cases} \tag{1.6}$$

The previous system represents a *regular perturbation problem*, because when $\varepsilon = 0$ the system remains an ordinary differential equations system.

Analogously to the previous computations, we seek for U and V of the form:

$$U(\sigma) = \sum_{n=0}^{+\infty} \varepsilon^n U_n(\sigma), \qquad V(\sigma) = \sum_{n=0}^{+\infty} \varepsilon^n V_n(\sigma)$$

Substituting such expressions in system (1.4) we obtain for $n = 0$:

$$\begin{cases} \dfrac{dU_0}{d\sigma} = 0 & U_0(0) = 1 \\ \dfrac{dV_0}{d\sigma} = -(K+U_0)V_0 + U_0 & V_0(0) = 0 \end{cases}$$

From the first equation: $U_0 = const$, in particular $U_0(\sigma) = U_0(0) = 1$ and substituting in the second equation we get $\frac{dV_0}{d\sigma} = -(K+1)V_0 + 1$, with $V_0(0) = 0$, from which

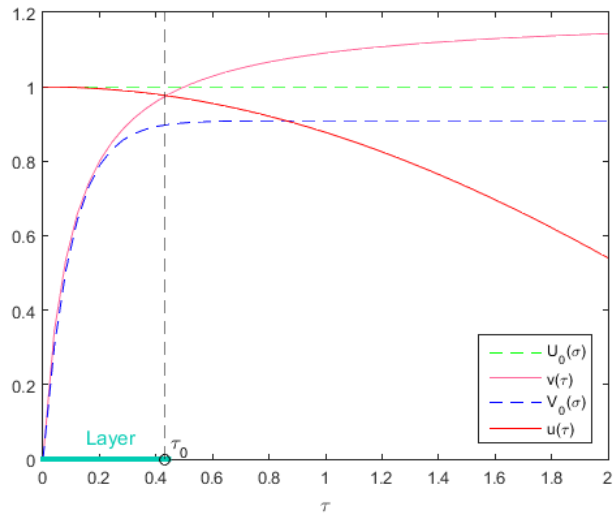
$$V_0(\sigma) = \frac{1 - e^{-(K+1)\sigma}}{K+1}$$

(U_0, V_0) is called *inner solution*, because it is a good approximation of (u,v) only inside the transient state.

1.2.4.4 Matching

We first verify that the following conditions hold true:

$$\lim_{\sigma \longrightarrow +\infty} U_0(\sigma) = \lim_{\tau \longrightarrow 0} u_0(\tau), \qquad \lim_{\sigma \longrightarrow +\infty} V_0(\sigma) = \lim_{\tau \longrightarrow 0} v_0(\tau).$$



Indeed

$$\begin{cases} \lim\limits_{\sigma \longrightarrow +\infty} U_0(\sigma) = \lim\limits_{\sigma \longrightarrow +\infty} 1 = 1 \\ \lim\limits_{\tau \longrightarrow 0} u_0(\tau) = u_0(0) = 1 \end{cases} =: L_u$$

$$\begin{cases} \lim\limits_{\sigma \longrightarrow +\infty} V_0(\sigma) = \lim\limits_{\sigma \longrightarrow +\infty} \dfrac{1}{K+1}(1 - e^{-(K+1)\sigma}) = \dfrac{1}{K+1} \\ \lim\limits_{\tau \longrightarrow 0} v_0(\tau) = \lim\limits_{\tau \longrightarrow 0} \dfrac{u_0(\tau)}{K + u_0(\tau)} = \dfrac{1}{K+1} \end{cases} =: L_v$$

We construct the ***uniform approximation*** as:

$$\begin{cases} \tilde{u}(\tau) = u_0(\tau) + U_0(\tau) - L_u = u_0(\tau) \\ \tilde{v}(\tau) = v_0(\tau) + V_0(\tau) - L_v = \dfrac{u_0(\tau)}{K + u_0(\tau)} - \dfrac{e^{-(K+1)\frac{\tau}{\varepsilon}}}{K+1} \end{cases}$$

It can be proved that $(\tilde{u}(\tau), \tilde{v}(\tau))$ satisfies

$$\begin{cases} |u(\tau) - \tilde{u}(\tau)| = O(\varepsilon) \\ |v(\tau) - \tilde{v}(\tau)| = O(\varepsilon) \end{cases} \qquad \forall \tau$$

Therefore, to compute the *uniform approximation* we have to solve the following system:

$$\begin{cases} \dfrac{d\tilde{u}}{d\tau} = -\tilde{u} + (\tilde{u}+\alpha)\dfrac{\tilde{u}}{K+\tilde{u}} = \dfrac{(\alpha-K)\tilde{u}}{K+\tilde{u}} \qquad \tilde{u}(0)=1 \\ \tilde{v}(\tau) = \dfrac{\tilde{u}(\tau)}{K+\tilde{u}(\tau)} - \dfrac{e^{-(K+1)\frac{\tau}{\varepsilon}}}{K+1} \end{cases} \tag{1.7}$$

We now derive the uniform approximation in dimensional form $(\tilde{s},\tilde{c})$. Substituting in system (1.7) the following expressions:

$$\frac{d\tilde{u}}{d\tau} = \frac{1}{k_1 e_0 s_0}\frac{d\tilde{s}}{dt}, \quad \tilde{u} = \frac{\tilde{s}(t)}{s_0}, \quad \tilde{v} = \frac{\tilde{c}(t)}{e_0}, \quad \alpha = \frac{k_{-1}}{k_1 s_0}, \quad K = \frac{k_{-1}+k_2}{k_1 s_0},$$

we obtain:

$$\begin{cases} \dfrac{1}{k_1 e_0 s_0}\dfrac{d\tilde{s}}{dt} = \left(\dfrac{\frac{k_{-1}}{k_1 s_0} - \frac{k_{-1}+k_2}{k_1 s_0}}{\frac{k_{-1}+k_2}{k_1 s_0} + \frac{\tilde{s}}{s_0}}\right)\dfrac{\tilde{s}(t)}{s_0} \\ \dfrac{\tilde{c}(t)}{e_0} = \dfrac{\frac{\tilde{s}(t)}{s_0}}{\frac{k_{-1}+k_2}{k_1 s_0} + \frac{\tilde{s}(t)}{s_0}} - \dfrac{1}{\frac{k_{-1}+k_2}{k_1 s_0}+1}\exp\left(-\left(\dfrac{k_{-1}+k_2}{k_1 s_0}+1\right)\dfrac{k_1 e_0 t}{\frac{e_0}{s_0}}\right). \end{cases}$$

After some computations and setting $k_m := \frac{k_{-1}+k_2}{k_1}$, we get:

$$\begin{cases} \dfrac{d\tilde{s}}{dt} = k_1 e_0 \tilde{s}(t)\left(\dfrac{-k_2}{k_{-1}+k_2+k_1\tilde{s}}\right) \\ \dfrac{\tilde{c}}{e_0} = \dfrac{\tilde{s}(t)}{k_m+\tilde{s}(t)} - \dfrac{s_0}{k_m+s_0}e^{-\left(\frac{k_m}{s_0}+1\right)k_1 s_0 t} \end{cases}$$

and finally:

$$\begin{cases} \dfrac{d\tilde{s}(t)}{dt} = -\dfrac{k_2 e_0 \tilde{s}}{k_m+\tilde{s}} \qquad \tilde{s}(0)=s_0 \\ \tilde{c}(t) = \dfrac{e_0\tilde{s}}{k_m+\tilde{s}} - \dfrac{e_0 s_0}{k_m+s_0}e^{-\left(\frac{k_m}{s_0}+1\right)k_1 s_0 t}. \end{cases}$$

1.3 Competitive inhibition

Competitive enzyme inhibition is a type of reaction in which a substance, called inhibitor (I), competes with the substrate to bind to the enzyme. The diagram of reaction is the following:

$$\mathrm{S} + \mathrm{E} \underset{k_{-1}}{\overset{k_1}{\rightleftarrows}} \mathrm{C}_1 \overset{k_2}{\rightarrow} \mathrm{P} + \mathrm{E}$$

$$\mathrm{I} + \mathrm{E} \underset{k_{-3}}{\overset{k_3}{\rightleftarrows}} \mathrm{C}_2$$

Denoting by i the concentration of I and applying the law of mass action, we obtain the ODE system:

$$\begin{cases} \dfrac{ds}{dt} = k_{-1}c_1 - k_1 es & s(0) = s_0 \\ \dfrac{de}{dt} = (k_{-1} + k_2)c_1 - k_1 es + k_{-3}c_2 - k_3 ei & e(0) = e_0 \\ \dfrac{di}{dt} = -k_3 ei + k_{-3}c_2 & i(0) = i_0 \\ \dfrac{dc_1}{dt} = k_1 es - (k_{-1} + k_2)c_1 & c_1(0) = 0 \\ \dfrac{dc_2}{dt} = k_3 ei - k_{-3}c_2 & c_2(0) = 0 \\ \dfrac{dp}{dt} = k_2 c & p(0) = 0 \end{cases} \tag{1.8}$$

The following *conservation properties* hold:

- ***Inhibitor conservation***:

$$\frac{di}{dt} + \frac{dc_2}{dt} = 0 \Rightarrow i + c_2 = i_0;$$

- ***Enzyme conservation***:

$$\frac{de}{dt} + \frac{dc_1}{dt} + \frac{dc_2}{dt} = 0 \Rightarrow e + c_1 + c_2 = e_0;$$

- ***Substrate conservation***:

$$\frac{ds}{dt} + \frac{dc_1}{dt} + \frac{dp}{dt} = 0 \Rightarrow s + c_1 + p = s_0.$$

By setting $e = e_0 - c_1 - c_2$ and eliminating the equation for the product p, the reduced system reads:

$$\begin{cases} \dfrac{ds}{dt} = -k_1 e_0 s + k_1 s c_2 + (k_1 s + k_{-1}) c_1 & s(0) = s_0 \\ \dfrac{di}{dt} = -k_3 e_0 i + k_3 i c_1 + (k_3 i + k_{-3}) c_2 & i(0) = i_0 \\ \dfrac{dc_1}{dt} = k_1 e_0 s - k_1 s c_2 - (k_1 s + k_{-1} + k_2) c_1 & c_1(0) = 0 \\ \dfrac{dc_2}{dt} = k_3 e_0 i - k_3 i c_1 - (k_3 i + k_{-3}) c_2 & c_2(0) = 0 \end{cases} \tag{1.9}$$

1.3.1 Quasi-steady state approximation

Assume that

$$\frac{dc_1}{dt} = \frac{dc_2}{dt} = 0.$$

Then, from the fourth equation of system (1.9), we get

$$c_2 = \frac{k_3 e_0 i - k_3 i c_1}{k_3 i + k_{-3}} = \frac{(e_0 - c_1) i}{i + K_i},$$

where $K_i = \frac{k_{-3}}{k_3}$. From the third equation in system (1.9), we obtain:

$$\begin{aligned} c_1 &= \frac{k_1 e_0 s - k_1 s c_2}{k_{-1} + k_2 + k_1 s} \\ &= \frac{k_1 \left(e_0 - \frac{(e_0 - c_1) i}{i + K_i} \right) s}{k_{-1} + k_2 + k_1 s} \\ &= \frac{k_1 s \left(\frac{e_0 K_i + c_1 i}{i + K_i} \right)}{k_{-1} + k_2 + k_1 s} \\ &= \frac{s(e_0 K_i + c_1 i)}{(K_m + s)(i + K_i)}, \end{aligned}$$

where $K_m = \frac{k_{-1} + k_2}{k_1}$. Then, after some computations we finally obtain:

$$c_1 = \frac{e_0 s}{s + K_m \left(1 + \frac{i}{K_i} \right)} \qquad \text{and} \qquad c_2 = \frac{K_m e_0 i}{K_i \left(s + K_m \left(1 + \frac{i}{K_i} \right) \right)}.$$

Thus, the reaction velocity in the quasi-steady state is given by:

$$v = \frac{dp}{dt} = k_2 c_1 = \frac{k_2 e_0 s}{s + K_m \left(1 + \frac{i}{K_i} \right)}.$$

1.4 Cooperativity

In some enzyme reactions, the velocity is not a Michaleis-Menten hyperbolic curve, but it presents a sigmoidal growth. This results from cooperative effects, where the enzyme can bind more than one substrate molecule but the binding of one substrate molecule affects the binding of subsequent ones, accelerating the process.

Suppose that the enzyme can bind two substrate molecules, so it can exist in one of the following three states:

- a free molecule E;
- a complex with one occupied binding site, C_1;
- a complex with two occupied binding sites, C_2.

The reaction diagram is:

$$\mathrm{S} + \mathrm{E} \underset{k_{-1}}{\overset{k_1}{\rightleftarrows}} \mathrm{C}_1 \overset{k_2}{\rightarrow} \mathrm{P} + \mathrm{E}$$

$$\mathrm{S} + \mathrm{C}_1 \underset{k_{-3}}{\overset{k_3}{\rightleftarrows}} \mathrm{C}_2 \overset{k_4}{\rightarrow} \mathrm{P} + \mathrm{C}_1$$

By applying the law of mass action, we obtain the ODE system:

$$\begin{cases} \dfrac{ds}{dt} = -k_1 es + k_{-1}c_1 - k_3 sc_1 + k_{-3}c_2 & s(0) = s_0 \\ \dfrac{de}{dt} = -k_1 es + (k_{-1} + k_2)c_1 & e(0) = e_0 \\ \dfrac{dc_1}{dt} = k_1 es - (k_{-1} + k_2)c_1 - k_3 sc_1 + (k_{-3} + k_4)c_2 & c_1(0) = 0 \\ \dfrac{dc_2}{dt} = k_3 sc_1 - (k_{-3} + k_4)c_2 & c_2(0) = 0 \\ \dfrac{dp}{dt} = k_2 c_1 + k_4 c_2 & p(0) = 0 \end{cases} \tag{1.10}$$

The following *conservation properties* hold:

- ***Enzyme conservation***:

$$\frac{de}{dt} + \frac{dc_1}{dt} + \frac{dc_2}{dt} = 0 \Rightarrow e + c_1 + c_2 = e_0;$$

- ***Substrate conservation***:

$$\frac{ds}{dt} + \frac{dc_1}{dt} + 2\frac{dc_2}{dt} + \frac{dp}{dt} = 0 \Rightarrow s + c_1 + 2c_2 + p = s_0.$$

By setting $e = e_0 - c_1 - c_2$ and eliminating the equation for the product p, the reduced system reads:

$$\begin{cases} \dfrac{ds}{dt} = -k_1 e_0 s + (k_1 - k_3)c_1 s + k_1 c_2 s + k_{-1} c_1 + k_{-3} c_2 & s(0) = s_0 \\ \dfrac{dc_1}{dt} = k_1 e_0 s - (k_1 + k_3)c_1 s - k_1 c_2 s - (k_{-1} + k_2)c_1 + (k_{-3} + k_4)c_2 & c_1(0) = 0 \\ \dfrac{dc_2}{dt} = k_3 c_1 s - (k_{-3} + k_4)c_2 & c_2(0) = 0 \end{cases} \tag{1.11}$$

1.4.1 Quasi-steady state approximation

Assume that

$$\frac{dc_1}{dt} = \frac{dc_2}{dt} = 0.$$

Then, from the third equation of system (1.11), we get

$$c_2 = \frac{k_3 c_1 s}{k_{-3} + k_4}.$$

Substituting the previous expression for c_2 into the second equation of (1.11) and after some computation we get

$$c_1 = \frac{K_c e_0 s}{K_c s + s^2 + K_m K_c} \qquad \text{and} \qquad c_2 = \frac{e_0 s^2}{K_c s + s^2 + K_m K_c},$$

where

$$K_m = \frac{k_{-1} + k_2}{k_1} \qquad \text{and} \qquad K_c = \frac{k_{-3} + k_4}{k_3}.$$

As a result, the reaction velocity in the quasi-steady state is given by:

$$v = \frac{dp}{dt} = k_2 c_1 + k_4 c_2 = \frac{(k_2 K_c + k_4 s)e_0 s}{K_c s + s^2 + K_m K_c}.$$

We now distinguish the three following cases:

- **no cooperativity.** The enzyme binding sites act independently and identically, then

  $$k_1 = 2k_3 = 2k_+, \qquad 2k_{-1} = k_{-3} = 2k_-, \qquad 2k_2 = k_4,$$

  where k_+ and k_- are the forward and backward reaction rates for the individual binding sites. The factors 2 occur because two identical binding sites are involved in the reaction, doubling the amount of the reactant. Thus the reaction velocity is given by the Michaelis-Menten formula

  $$v = \frac{2k_2 e_0 (K + s)s}{K^2 + 2Ks + s^2} = \frac{2k_2 e_0 s}{K + s},$$

where $K = \frac{k_- + k_2}{k_+}$.

- **positive cooperativity**. Suppose that the binding of the first substrate molecule is slow, but that, with one site bound, binding of the second is fast. This can be modeled by letting
$$k_3 \to +\infty \qquad \text{and } k_1 \to 0,$$
while keeping $k_1 k_3$ constant. As a consequence
$$K_c \to 0 \qquad \text{and } K_m \to +\infty,$$
whith $K_m K_c$ constant. In this case, the reaction velocity is given by
$$v = \frac{k_4 e_0 s^2}{s^2 + K_m K_c}. \tag{1.12}$$
(1.12) is known as *Hill equation*.
- **negative cooperativity**. In this situation the binding of the first substrate molecule decreases the rate of subsequent binding. This can be modeled by letting
$$k_3 \to 0 \qquad \text{and } k_1 \to +\infty,$$
while keeping $k_1 k_3$ constant. As a consequence
$$K_c \to +\infty \qquad \text{and } K_m \to 0,$$
whith $K_m K_c$ constant. In this case, the reaction velocity is given by the Michaelis-Menten formula
$$v = \frac{k_2 e_0 s}{s + K_m}. \tag{1.13}$$

Figure 1.3 reports an example of behavior of the reaction velocity in the three cases.

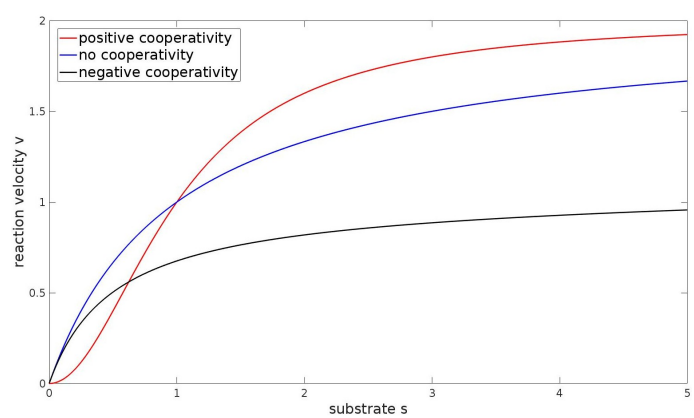


Fig. 1.3: Cooperativity: behavior of reaction velocity.

Chapter 2
Electrodiffusion models

In this chapter, we briefly review the main mathematical models used in cellular electrophysiology, and we refer to [17] for a more complete treatment and additional topics.

2.1 Excitable cellular membranes

A fundamental topic in molecular biophysics is the study of ionic currents through protein channels of excitable cellular membranes as a function of the ionic concentrations, applied voltage and channel structure.

The cellular membrane is a lipid bilayer in which are immersed proteins. The membrane contains water-filled pores and protein-lined pores, called *ionic channels*, which allow the flow of specific ions, primarily Na^+, K^+, Cl^-, and Ca^{2+}. The membrane acts as a barrier to the free flow of ions and maintains concentration differences of these ions. The concentration gradients produce a potential difference across the membrane, the *transmembrane potential*, that drives the ionic currents.

Concentration differences are set up and maintained by active transport mechanisms that use energy to pump ions against their concentration gradients.

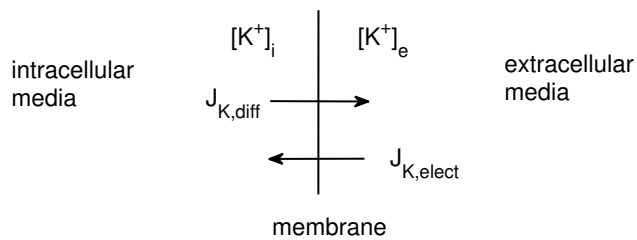
One of the most important of these pumps is the $Na^+ - K^+$ $ATPase$ sodium - potassium pump: it uses energy stored in ATP (adenosine triphosphate) molecules to pump Na^+ out of the cell and K^+ in

$$ATP + 3[Na^+]_i + 2[K^+]_e \rightarrow ADP + P_i + 3[Na^+]_e + 2[K^+]_i,$$

releasing ADP (adenosine diphosphate) and inorganic phosphate P_i. For example, in the squid giant axon: $[Na^+]_i = 50\ mM\ <\ [Na^+]_e = 437\ mM$, $[K^+]_i = 397\ mM\ >\ [K^+]_e = 20\ mM$. At the equilibrium, this biochemical pump maintains a potential jump between the intra- and extracellular space of $v_{rest} = -65\ mV$.

2.2 The Nernst-Planck equation

The ionic concentration gradients maintained across the membrane yield a jump of electric potential. The equation that describes this process is the Nernst-Planck equation.



In general, the flux across the membrane of the generic ion K with valence z is the sum of two contributions, the diffusion flux J_{diff} and the electric flux J_{elect}, all measured in $mol\ cm^{-2}\,s^{-1}$.

Due to the ion motion by diffusion, down the concentration gradient, the diffusion flux J_{diff} satisfies *Fick law*:

$$J_{diff} = -D\nabla c,$$

where D is the diffusion coefficient ($cm^2\,s^{-1}$) and c is the concentration ($mol\ cm^{-3}$).

Due to the ion motion by electric field, the electric flux J_{elect} satisfies the *Planck equation*:

$$J_{elect} = -\frac{z}{|z|}\mu c\nabla u,$$

where u is the electric potential (V), μ is the mobility of the ion ($cm^2\,V^{-1}\,s^{-1}$), z is the valence of the ion, so that $\frac{z}{|z|} = \begin{cases} +1 \text{ for positive ions} \\ -1 \text{ for negative ions.} \end{cases}$

We note that if the concentration gradient is zero then $J_{diff} = 0$, while if the potential gradient is zero then $J_{elect} = 0$.

In both the electric and the diffusion fluxes, the ions collide with the solvent and therefore the diffusion coefficient D and the ion mobility μ should be related. Indeed, Einstein [10] found that this relationship is given by

$$D = \frac{\mu RT}{|z|F}, \tag{2.1}$$

where R is the gas constant $(8.314\,J/(K\cdot mol))$, T the absolute temperature $(Kelvin)$, $F = eN_A$ the Faraday constant $(9.6485e+4\ Coulomb/mole)$, e the elementary charge and N_A the Avogadro number. Then it follows that the total ionic flux J is given by

$$J = J_{diff} + J_{elect} = -D\nabla c - \frac{z}{|z|}\mu c \nabla u,$$

hence, using (2.1), we obtain the *Nernst-Planck equation*

$$J = -D\left(\nabla c + \frac{zF}{RT} c \nabla u\right). \tag{2.2}$$

Let us assume now that the ion flux J and the electric potential u are transverse to the cell membrane, which extends from $x = 0$ (inside) to $x = L$ (outside), and the diffusion coefficient D is constant. Moreover, if we are at the steady state $\partial_t c = 0$, thus the conservation law

$$\partial_t c + \partial_x J = 0$$

implies that J is constant. Hence, we can reduce equation (2.2) to the one-dimensional problem

$$\frac{dc}{dx}(x) + \frac{zF}{RT}\frac{du}{dx}(x)c(x) + \frac{J}{D} = 0. \tag{2.3}$$

Equation (2.3) is clearly a linear differential equation in $c(x)$, whose solution can be obtained in terms of $u(x)$ and J by multiplying by

$$\exp\left(\frac{zF}{RT}\int_0^x \frac{du}{ds}(s)ds\right)$$

and integrating on $(0,x)$, yielding

$$c(x) = \exp\left(-\frac{zF}{RT}(u(x) - u(0))\right)\left(c(0) - \frac{J}{D}\int_0^x \exp\left(\frac{zF}{RT}(u(s) - u(0))\right)ds\right). \tag{2.4}$$

2.3 The Goldman-Hodgkin-Katz (GHK) current-voltage relation

Suppose the electric field constant across the membrane, then

$$\frac{du}{dx} = -\frac{v}{L},$$

where $v = u(0) - u(L) = u_i - u_e$ and we have adopted the standard notation with indexes i, e denoting intra- and extracellular quantities. In this case, setting $c(0) = c^i$, the solution (2.4) of the Nernst-Planck equation is

$$c(x) = \frac{JRTL}{DzvF}\left(1 - \exp\left(\frac{zvF}{RTL}x\right)\right) + c^i \exp\left(\frac{zvF}{RTL}x\right).$$

In order to satisfy $c(L) = c^e$, it must be that

$$J = \frac{DzFv}{LRT} \frac{c^i - c^e \exp\left(\frac{-zvF}{RT}\right)}{1 - \exp\left(\frac{-zvF}{RT}\right)}.$$

The *electrical current density* in $C\ s^{-1} cm^{-2} = A\ cm^{-2}$ is defined by $I := zFJ$, hence we obtain the *Goldman-Hodgkin-Katz (GHK)* current-voltage relation

$$I = P\frac{z^2F^2}{RT} v \frac{c^i - c^e \exp\left(\frac{-zvF}{RT}\right)}{1 - \exp\left(\frac{-zvF}{RT}\right)}, \tag{2.5}$$

where $P = \frac{D}{L}$ is the permeability of the membrane to the ion K.

The GHK flux equation is applied in several fields, one of the most important being in modeling the L-type channels in calcium dynamics, see e.g. [4].

2.4 Nernst equilibrium potential

We consider a membrane separating two neutral solutions with different ionic concentrations, e.g. the salt KCl composed by the ions potassium K^+ and chloride Cl^-. Moreover, we suppose that the membrane is permeable only to one of the two ions, that we denote by K, with generic valence z.

At thermodynamical equilibrium, each local process and its reverse proceed at the same rate, hence $J = 0$ and the solution (2.4) of the Nernst-Planck equation becomes

$$c(x) = \exp\left(-\frac{zF}{RT}(u(x) - u(0))\right) c(0),$$

from which follows in $x = L$

$$\log\left(\frac{c(L)}{c(0)}\right) = -\frac{zF}{RT}\left(u(L) - u(0)\right).$$

Using the standard notation with indexes i, e denoting intra- and extracellular quantities, respectively, we then obtain the *Nernst equation* for the equilibrium (reversal) potential

$$v_K := u_i - u_e = -\frac{RT}{zF} \log\left(\frac{c^i}{c^e}\right).$$

This derivation is based on the constitutive Nernst - Plank law for the total flux. For a thermodynamical derivation from electrochemical potentials that shows the universal character of the Nernst potential, see the next Section.

For ventricular myocytes, the typical intra- and extracellular concentration and Nernst potential values are given in Table 2.1.

	extracellular concentration (mM)	intracellular concentration (mM)	Nernst potential (mV)
Na^+	145	15	60
Cl^-	100	5	-80
K^+	4.5	160	-95
Ca^{2+}	1.8	1e-4	130
H^+	1e-4	2e-4	-18

Table 2.1: Intra- and extracellular concentration and Nernst potential values for ventricular myocytes.

2.5 Electrodiffusion models: the Poisson-Nernst-Planck (PNP) equation

We now suppose to have two types of ions, a cation S_1 and an anion S_2, with concentrations c_1 and c_2, valences $z_1 = z > 0$ and $z_2 = -z < 0$. We consider the flux through a channel H connecting the intra- and extracellular neutral spaces. The Nernst-Planck equations for the fluxes J_1, J_2 of S_1 and S_2 state that

$$J_1 = -D_1 \left(\nabla c_1 + \frac{zF}{RT} c_1 \nabla u \right),$$
$$J_2 = -D_2 \left(\nabla c_2 - \frac{zF}{RT} c_2 \nabla u \right),$$

and the conservation law (continuity equation)

$$\partial_t c_1 + \operatorname{div} J_1 = 0 \quad \text{in } H,$$
$$\partial_t c_2 + \operatorname{div} J_2 = 0 \quad \text{in } H.$$

Furthermore, by Gauss law

$$\operatorname{div} D = \rho \quad \text{in } H,$$

where $D = \varepsilon_a E$ is the dielectric displacement associated to the electric field E, ε_a the dielectric constant in the ionic solution, ρ the charge density. Since $E = -\nabla u$ with u the electric potential, we have $D = -\varepsilon_a \nabla u$ and since in one mole there are $qN_A z = Fz$ charges, we have $\rho = N_a q(z_1 c_1 + z_2 c_2) = Fz(c_1 - c_2)$. Hence, inside the channel H

$$-\Delta u = \frac{Fz}{\varepsilon_a}(c_1 - c_2).$$

We denote by S the lateral surface of the channel H and by $\partial H_i, \partial H_e$ the two channel openings in contact with the intra and extracellular media, respectively. Then, we make the following additional assumptions, yielding boundary conditions for the ionic concentrations, fluxes and the electric potential:

a) the cation and anion concentrations on both sides of the membrane are constant, so that the intra- and extracellular spaces are in electrical equilibrium

$$c_1(x,t)|_{\partial H_i} = c_2(x,t)|_{\partial H_i} = c^i, \quad c_1(x,t)|_{\partial H_e} = c_2(x,t)|_{\partial H_e} = c^e;$$

b) since the dielectric constants ε_a of the channel's solution and ε_m of the membrane satisfy $\varepsilon_m << \varepsilon_a$, we assume that the channel wall is electrically insulated,

$$n^T E|_S = 0, \text{ i.e. } n^T \nabla u|_S = 0;$$

c) the channel wall prevents the movement of ions toward the lipidic regions,

$$n^T J_i|_S = 0, \quad n^T J_e|_S = 0;$$

d) the electric potentials at the channel openings in contact with the extracellular and intracellular media are constant and equal to u_i and u_e, respectively, but, since the potential is given up to an additive constant, we set

$$u|_{\partial H_i} = u_i - u_e = v, \quad u|_{\partial H_e} = 0.$$

The *Poisson-Nernst-Planck* (PNP) equation treats discrete ions in a channel as a continuum charge distribution, representing the average ion fluxes in terms of densities and potential gradients. For a derivation of the PNP system from the Langevin model of ionic motion, see Schuss et al. [26]. For a singular perturbation analysis of the steady-state PNP system, see e.g. [3, 28] and the references therein. In addition to continuum PNP equations, more detailed approaches have been studied, such as all-atom Molecular Dynamics (MD), non-linear Poisson-Boltzmann (PB) equation, Brownian Dynamics (BD), see Roux et al. [24]. For a multiscale model linking MD and electrophysiology for the cardiac I_{Ks} channel, see Silva et al. [27].

Since the channel has a diameter of a few Angstroms, it can be assumed cylindrical and the membrane around the channel planar. Then the ionic concentrations c_1, c_2, fluxes J_1, J_2 and potential u are constant on each cross section of the cylindrical channel H and depend only on the position x across the cylindrical channel H, that we assume to range from 0 (intracellular media) to L (extracellular media).

We now consider the steady-state of the 1-D model, assuming that the ion concentrations are constant across the channel section. Collecting the previous equations and boundary conditions, we find the one-dimensional *Poisson-Nernst-Plank* (PNP) system:

$$\begin{cases} \partial_t c_1 + \partial_x J_1 = 0 \ \text{ with } \ J_1 = -D_1 \left(\partial_x c_1 + \dfrac{zF}{RT} c_1 \partial_x u \right) \\ \partial_t c_2 + \partial_x J_2 = 0 \ \text{ with } \ J_2 = -D_2 \left(\partial_x c_2 - \dfrac{zF}{RT} c_2 \partial_x u \right) \\ \partial_{xx}^2 u = -\dfrac{zF}{\varepsilon_a}(c_1 - c_2) \\ \text{ with boundary conditions:} \\ c_1(0,t) = c_2(0,t) = c^i \quad c_1(L,t) = c_2(L,t) = c^e \\ u(0,t) = v \quad u(L,t) = 0 \\ \text{ and initial conditions:} \\ u(x,0) = 0 \quad c_1(x,0) = c_2(x,0) = 0. \end{cases}$$

At steady state $\partial_t c_1 = \partial_t c_2 = 0$, hence the stationary PNP system becomes

$$\begin{cases} \dfrac{d}{dx} D_1 \left(\dfrac{dc_1}{dx} + \dfrac{zF}{RT} c_1 \dfrac{du}{dx} \right) = 0, \\ \dfrac{d}{dx} D_2 \left(\dfrac{dc_2}{dx} - \dfrac{zF}{RT} c_2 \dfrac{du}{dx} \right) = 0, \\ \dfrac{d^2 u}{dx^2} = -\dfrac{zF}{\varepsilon_a}(c_1 - c_2) \\ c_1(0) = c_2(0) = c^i \\ c_1(L) = c_2(L) = c^e \\ u(0) = v \quad u(L) = 0. \end{cases} \tag{2.6}$$

This one-dimensional non-linear boundary value system admits a unique solution $c_1(x),\ c_2(x),\ u(x)$, see [22] and for a perturbation analysis [3]. Multidimensional existence results follow from the theory of semiconductor equations, see e.g. [18], and exploiting the gradient flow structure of the PNP system, see e.g. [2]. The system can be written in dimensionless form by rescaling the variables and functions as

$$\begin{aligned} & y = x/L, \quad \phi = uzF/RT, \quad p = \frac{c_1}{c^i + c^e}, \quad n = \frac{c_2}{c^i + c^e}, \\ & J_p = J_1 \frac{L}{(c^i + c^e) D_1}, \quad J_n = J_2 \frac{L}{(c^i + c^e) D_2}, \quad \hat{c}^{i,e} = \frac{c^{i,e}}{c_i + c_e}, \quad \hat{v} = \frac{vzF}{RT}, \end{aligned}$$

so that

$$\begin{aligned} & \frac{d^2\phi}{dy^2} = \frac{zFL^2}{RT} \frac{d^2 u}{dx^2} = -\frac{z^2 F^2 L^2}{RT \varepsilon_a} (c^i + c^e)(p - n), \\ & -J_p = \frac{dp}{dy} + p \frac{d\phi}{dy}, \\ & -J_n = \frac{dn}{dy} - n \frac{d\phi}{dy}. \end{aligned}$$

The first two equations of (2.6) require that $\dfrac{dJ_p}{dy} = \dfrac{dJ_n}{dy} = 0$, i.e. that J_p and J_n are constant. Therefore, the dimensionless one-dimensional stationary PNP system becomes:

find $\{p(x), n(x), \phi(x)\}$ and constants $\{J_p, J_n\}$ such that

$$\begin{cases} -J_p = \dfrac{dp}{dy} + p\dfrac{d\phi}{dy}, \\ -J_n = \dfrac{dn}{dy} - n\dfrac{d\phi}{dy} \\ \dfrac{d^2\phi}{dy^2} = -\lambda^2(p-n), \\ \text{with boundary conditions:} \\ p(0) = n(0) = \hat{c}^i, \quad p(1) = n(1) = \hat{c}^e, \quad \phi(0) = \hat{v}, \quad \phi(1) = 0, \end{cases} \tag{2.7}$$

with $\lambda^2 = \dfrac{z^2F^2L^2}{\varepsilon_a RT}(c^i + c^e)$.

Since no analytical solutions of this system are known, we will look for approximate solutions obtained by perturbation techniques with respect to the parameter λ.

2.5.1 PNP: the short channel or low concentrations limit

We look first for approximate solutions when the parameter $\lambda = \dfrac{zF}{\sqrt{\varepsilon_a RT}} L\sqrt{c^i + c^e}$ is small, corresponding to *short channels* or *low ionic concentrations*. This is a regular perturbation problem and in order to obtain an approximation with error of order $O(\lambda)$ we solve the reduced problem in the limit case of $\lambda = 0$. Thus

$$\frac{d^2\phi}{dy^2} = 0$$

implies that the electric field $E = -\dfrac{d\phi}{dy}$ is constant and the potential $\phi(y)$ is linear, so from the boundary conditions we obtain

$$\phi(y) = \hat{v} - \hat{v}y, \quad \frac{d\phi}{dy} = -\hat{v}.$$

Then the differential equation for p becomes

$$-J_p = \frac{dp}{dy} - \hat{v}p.$$

Integrating between 0 and y and using the left boundary condition $p(0) = \hat{c}^i$, we obtain

$$p(y) = \hat{c}^i e^{\hat{v}y} + J_p \frac{1 - e^{\hat{v}y}}{\hat{v}}.$$

The right boundary condition $p(1) = \hat{c}^e$ then determines

$$J_p = \hat{v}\frac{\hat{c}^i - \hat{c}^e e^{-\hat{v}}}{1 - e^{-\hat{v}}}.$$

Analogously, the differential equation for n becomes

$$-J_n = \frac{dn}{dy} + \hat{v}n,$$

so, by integrating and imposing the boundary conditions, we obtain

$$n(y) = \hat{c}^i e^{-\hat{v}y} - J_n \frac{1 - e^{-\hat{v}y}}{\hat{v}},$$

$$J_n = -\hat{v}\frac{\hat{c}^i - \hat{c}^e e^{\hat{v}}}{1 - e^{\hat{v}}}.$$

Returning to the original variables, we then determine the dimensional fluxes J_1, J_2, hence the ionic density currents

$$\begin{aligned} I_1 = zFJ_1 &= \frac{z^2F^2}{RT}\frac{D_1}{L}v\frac{c^i - c^e e^{-\frac{zF}{RT}v}}{1 - e^{-\frac{zF}{RT}v}}, \\ I_2 = zFJ_2 &= \frac{z^2F^2}{RT}\frac{D_2}{L}(-v)\frac{c^i - c^e e^{\frac{zF}{RT}v}}{1 - e^{\frac{zF}{RT}v}}. \end{aligned}$$

These are GHK current-voltage relations (2.5), with $P_K = D_K/L, \;\; K = 1, 2$.

2.5.2 PNP: the long channel or high concentrations limit

We now look for approximate solutions when the parameter λ is large. We define the small parameter $\eta = 1/\lambda = \dfrac{\sqrt{\varepsilon_a RT}}{zF}\dfrac{1}{L\sqrt{c^i + c^e}}$ and consider the singular perturbation problem for the potential

$$\eta^2 \frac{d^2\phi}{dy^2} = n - p.$$

The reduced problem obtained in the limit case for $\eta = 0$ implies that $p(y) = n(y)$, so by adding or subtracting the equations for J_p and J_n in (2.7) we have

$$\begin{cases} -J_p - J_n = \dfrac{dp}{dy} + \dfrac{dn}{dy} = 2\dfrac{dp}{dy}, \\ -J_p + J_n = p\dfrac{d\phi}{dy} + n\dfrac{d\phi}{dy} = 2p\dfrac{d\phi}{dy} \\ p(0) = n(0) = \hat{c}^i, \quad p(1) = n(1) = \hat{c}^e, \quad \phi(0) = \hat{v}, \quad \phi(1) = 0. \end{cases} \tag{2.8}$$

The first equation requires $\dfrac{dp}{dy}$ to be constant and from the boundary conditions we obtain $p(y) = \hat{c}^i + (\hat{c}^e - \hat{c}^i)y$, therefore system (2.8) becomes

$$\begin{cases} \dfrac{J_p + J_n}{2} = (\hat{c}^e - \hat{c}^i) \\ \dfrac{d\phi}{dy} = \dfrac{\hat{J}}{p(y)} = \dfrac{\hat{J}}{\hat{c}^i + (\hat{c}^e - \hat{c}^i)y}, \end{cases} \tag{2.9}$$

with $\hat{J} = (J_n - J_p)/2$, yielding

$$\phi(y) = \hat{v} + \frac{\hat{J}}{\hat{c}^e - \hat{c}^i} \log \frac{\hat{c}^i + (\hat{c}^e - \hat{c}^i)y}{\hat{c}^i}.$$

Then the boundary condition $\phi(1) = 0$ gives us the condition $\hat{v} + \dfrac{\hat{J}}{\hat{c}^e - \hat{c}^i} \log \dfrac{\hat{c}^e}{\hat{c}^i} = 0$, that determines $\hat{J} = (\hat{c}^e - \hat{c}^i)\dfrac{-\hat{v}}{\log \frac{\hat{c}^e}{\hat{c}^i}}$, which coupled with (2.9) implies

$$\begin{cases} -J_p = (\hat{c}^e - \hat{c}^i)(\dfrac{-\hat{v}}{\log \frac{\hat{c}^e}{\hat{c}^i}} + 1), \\ J_n = (\hat{c}^e - \hat{c}^i)(\dfrac{-\hat{v}}{\log \frac{\hat{c}^e}{\hat{c}^i}} - 1). \end{cases}$$

Returning to the dimensional fluxes $J_1 = \dfrac{c^i + c^e}{L} D_1 J_p, \;\; J_2 = \dfrac{c^i + c^e}{L} D_2 J_n$, we find that the dimensional currents $I_1 = zF\dfrac{c^i + c^e}{L} D_1 J_p, \;\; I_2 = -zF\dfrac{c^i + c^e}{L} D_2 J_n$ are

$$\begin{aligned} I_1 &= \frac{zF}{L}(c^e - c^i)D_1 \left(\frac{v}{\frac{RT}{zF} \log \frac{c^e}{c^i}} - 1 \right) = \frac{zFD_1}{L} \frac{c^e - c^i}{v_1}(v - v_1), \\ I_2 &= -\frac{zF}{L}(c^e - c^i)D_2 \left(\frac{v}{\frac{RT}{zF} \log \frac{c^e}{c^i}} - 1 \right) = \frac{-zFD_2}{L} \frac{c^e - c^i}{v_2}(v - v_2), \end{aligned} \tag{2.10}$$

where $v_k = \dfrac{RT}{z_k F} \log \dfrac{c^e}{c^i}$, $k = 1, 2$ are the Nernst potentials of the two ions considered.

These linear current-voltage relations are derived using the approximation of a constant diffusive gradient.

2.5.3 Equilibrium potential for multi-ion fluxes

The Nernst equilibrium potential has a derivation based on universal thermodynamic principles only in the simpler case of a single-ion flux. If we consider two or more

ions, the situation is more complex and in general there is no potential that zeros each individual ionic current. However, we can still compute a reverse potential that zeros the total ionic current.

For example, if we consider the two ions Na^+, K^+ and we assume that their channels obey GHK current-voltage relations

$$I_{Na^+} = \frac{F^2}{RT} v P_{Na} \frac{c^i_{Na} - c^e_{Na} e^{-\frac{F}{RT}v}}{1 - e^{-\frac{F}{RT}v}},$$
$$I_{K^+} = \frac{F^2}{RT} v P_K \frac{c^i_K - c^e_K e^{-\frac{F}{RT}v}}{1 - e^{-\frac{F}{RT}v}},$$

with $P_{Na} = \frac{D_{Na}}{L}, P_K = \frac{D_K}{L}$ and the reverse potential at which $I_{Na^+} + I_{K^+} = 0$ is

$$v_r = -\frac{RT}{F} \log \frac{P_{Na} c^i_{Na} + P_K c^i_K}{P_{Na} c^e_{Na} + P_K c^e_K}.$$

If instead the channels obey linear current - voltage relations

$$I_{Na^+} = G_{Na}(v - v_{Na}), \quad I_{K^+} = G_K(v - v_K),$$

then the reverse potential that zeros the total current is

$$v_r = \frac{G_{Na} v_{Na} + G_K v_K}{G_{Na} + G_K}.$$

Analogously, if we consider the three ions Na^+, K^+, Cl^- with GHK channels, we have

$$v_r = -\frac{RT}{F} \log \frac{P_{Na} c^i_{Na} + P_K c^i_K + P_{Cl} c^i_{Cl}}{P_{Na} c^e_{Na} + P_K c^e_K + P_{Cl} c^e_{Cl}},$$

while with linear channels we have

$$v_r = \frac{G_{Na} v_{Na} + G_K v_K + G_{Cl} v_{Cl}}{G_{Na} + G_K + G_{Cl}}.$$

Chapter 3
Mathematical models of the cellular membrane

3.1 Electrical circuit model of the cellular membrane

Since the cell membrane separates charges that accumulate at its intra- and extracellular surfaces, it can be viewed as a capacitor. The capacitance is defined as the ratio between the charge Q across the capacitor and the voltage potential drop v necessary to hold the charge

$$C_m = \frac{Q}{v}.$$

Since the capacitive current is $I_{cap} = dQ/dt$, if C_m is constant, we have

$$I_{cap} = \frac{dQ}{dt} = C_m \frac{dv}{dt}.$$

The cellular membrane is modeled as a capacitor in parallel with a resistor (ionic current), as shown in Fig. 3.1, so by the current conservation law, the transmembrane current given by the sum of the capacitative and ionic currents, must be equal to the applied current I_{app}

$$C_m \frac{dv}{dt} + I_{ion} = I_{app}. \tag{3.1}$$

The structure of the total ionic current will be described by the specific ionic membrane model adopted. In order to discuss these models, we need to introduce the formalism for modeling ion channel gating.

3.2 Ion channel gating

In general, for a given transmembrane potential drop, the ionic current through a population of ionic channels is given in a unit area of membrane surface and can be

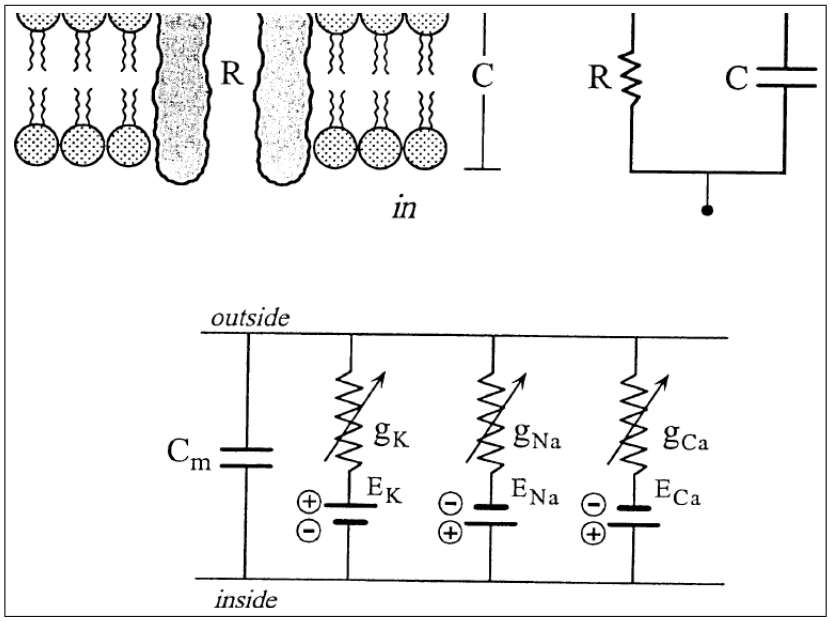


Fig. 3.1: Electrical circuit model of the cellular membrane.

modeled as a product

$$I_{ion} = g(v,t)\phi(v),$$

where $g(v,t)$ is the proportion of open channels in a unit area of the membrane surface and $\phi(v)$ is the current-voltage (I-V) relation of a single open channel. In the previous sections on electrodiffusion models, we have seen different models for $\phi(v)$. For example, in the long channel limit (2.10), we have $\phi(v) = g_c(v - v_r)$, where g_c and v_r are the channel conductance and resting potential, respectively. The proportion of open channels in a unit area of the membrane surface can be written as $g(v,t) = \dfrac{N}{S} = \dfrac{N_{tot}}{S}\dfrac{N}{N_{tot}}$, where N is the number of open channels on the membrane surface, N_{tot} is the total number of membrane channels, S is the membrane surface area. Hence, in the long channel limit we have $I_{ion} = \overline{G}_c w(v - v_r)$, where $\overline{G}_c = \dfrac{N_{tot} g_c}{S}$ is the maximal channel conductance per unit area of the membrane surface and $w = \dfrac{N}{N_{tot}}$ is the percentage of open channels. We now show how to model this percentage w, also known as gating variable, starting with the simplest case of a single-unit two-state channel.

One unit protein with two states. Consider a membrane portion of unit area containing a given type of ionic channels that behave independently and that assume two states S_0 (close) or S_1 (open), with transition rate constants α, β (in general

dependent on the transmembrane potential v) as indicated in the first order kinetic diagram

$$\text{close state} = S_0 \underset{\beta}{\overset{\alpha}{\rightleftarrows}} S_1 = \text{open state}.$$

If $S_0(t), S_1(t)$ are the average numbers of channels that at time t are in the state S_0, S_1, respectively, then

$$\begin{cases} \dfrac{dS_1(t)}{dt} = \alpha\, S_0(t) - \beta\, S_1(t) \\ S_1(t) + S_0(t) = S. \end{cases}$$

By setting $s_0 = S_0(t)/S, s_1 = S_1(t)/S$, the percentages of close and open channels per unit area of the surface membrane, we then have

$$\begin{cases} \dfrac{ds_1}{dt} = \alpha\, s_0 - \beta\, s_1 \\ s_0 + s_1 = 1. \end{cases}$$

Eliminating s_0 and denoting by $w = s_1$ the gating variable, we have

$$\frac{dw}{dt} = \alpha\,(1-w) - \beta\, w\,.$$

By setting $w_\infty = \frac{\alpha}{\alpha+\beta}$ and $\tau_w = \frac{1}{\alpha+\beta}$, this equation can be rewritten as

$$\frac{dw}{dt} = \frac{w_\infty - w}{\tau_w}, \tag{3.2}$$

where w_∞ and τ_w are the equilibrium state and the time constant, respectively.

By voltage-clamp or patch-clamp techniques, it is possible to apply an external current that balances the membrane current in such a way that the transmembrane potential v is kept at a fixed value in time. In this case, the capacitative current $I_{cap} = C_m \frac{dv}{dt}$ is zero and in the circuit model equation (3.1) in the long-channel limit we have $I_{ion}(t,v) = \overline{G}\, w(t,v)\,(v - v_r) = I_{app}$, hence the membrane conductance can be obtained as

$$\overline{G}\, w(t,v) = \frac{I_{app}(t,v)}{v - v_r}.$$

For each fixed value of v, these experimental curves can be plotted in time and compared with the theoretical curves predicted by solving the gating equation (3.2) in the voltage-clamp setting: if at time $t = 0$ v is increased from 0 to v_0 and held fixed at such value, the the time behavior is given by

$$w(t) = w_\infty(v_0)(1 - e^{-t/\tau_w(v_0)}), \tag{3.3}$$

while if at time $t = T$ we return v back to 0, we obtain the evolution

$$w(t) = w_\infty(v_0)e^{-(t-T)/\tau_w(v_0)}), \tag{3.4}$$

see Fig. 3.2. If the experimental conductance curves do not agree with these exponential curves, then the single-unit two-state channel model is not adequate and more complex models must be considered. Such models are introduced in the next sections and indeed they will be needed to correctly describe the ionic currents in the membrane models developed in the literature, starting with the classical Hodgkin-Huxley model presented in Sec. (3.3.1).

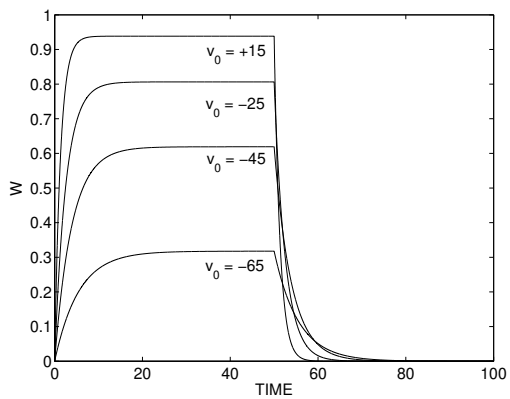


Fig. 3.2: Plots of voltage-clamp solutions (3.3) on the interval $[0\ T]$ and (3.4) on $[T\ 100]$, with $T = 50\ ms$, for $v_0 = -65, -45, -25, 15\ mV$.

Two subunits with two states. Analogously, if we consider ionic channels with 2 equal and independent subunits, which can open and close according to the following kinetic diagram

$$\text{close state } S_0 \underset{\beta}{\overset{2\alpha}{\rightleftarrows}} S_1 \underset{2\beta}{\overset{\alpha}{\rightleftarrows}} S_2 = \text{open state},$$

we have

$$\begin{cases} \dfrac{ds_0}{dt} = \beta s_1 - 2\alpha s_0 \\ \dfrac{ds_1}{dt} = -\beta s_1 + 2\alpha s_0 - \alpha s_1 + 2\beta s_2 \\ \dfrac{ds_2}{dt} = \alpha s_1 - 2\beta s_2. \end{cases}$$

From the condition $\frac{d}{dt}(s_0 + s_1 + s_2) = 0$ it follows $s_0 + s_1 + s_2 = 1$, hence

$$\begin{cases} \dfrac{ds_0}{dt} = \beta s_1 - 2\alpha s_0 \\ \dfrac{ds_2}{dt} = \alpha s_1 - 2\beta s_2 \\ s_1 = 1 - s_0 - s_2, \end{cases}$$

and after a change of variable

$$s_2 = n^2, \quad s_1 = 2n(1-n), \quad s_0 = (1-n)^2$$
$$\frac{dn}{dt} = \alpha(1-n) - \beta n.$$

k subunits with two states. Consider now an ionic channel with k equal and independent subunits, which can open and close according to the following kinetic diagram

$$\begin{tikzcd} S_0 \arrow[r, shift left, "k\,\alpha"] & S_1 \arrow[l, shift left, "\beta"] \arrow[r, shift left, "(k-1)\,\alpha"] & S_2 \arrow[l, shift left, "2\,\beta"] & \cdots & S_{k-1} \arrow[r, shift left, "\alpha"] & S_k \arrow[l, shift left, "k\,\beta"] \\ s_0 & s_1 & s_2 & & s_{k-1} & s_k \end{tikzcd}$$

As in the case with 2 subunits, after a change of variable, we obtain

$$s_2 = n^k, \quad s_j = \binom{k}{j} n^j (1-n)^{k-j}$$
$$\frac{dn}{dt} = \alpha(1-n) - \beta n.$$

Two different subunits. We consider now two subunits of different type, that we will denote by m and h. The transition rate constants are α, β for m and γ, δ for h, as indicated in the following kinetic diagram

$$\begin{tikzcd} S_{00} \arrow[r, shift left, "2\,\alpha"] \arrow[d, shift right, "\gamma"'] & S_{10} \arrow[l, shift left, "\beta"] \arrow[r, shift left, "\alpha"] \arrow[d, shift right, "\gamma"'] & S_{20} \arrow[l, shift left, "2\,\beta"] \arrow[d, shift right, "\gamma"'] \\ S_{01} \arrow[r, shift left, "\alpha"] \arrow[u, shift right, "\delta"'] & S_{11} \arrow[l, shift left, "\beta"] \arrow[r, shift left, "\alpha"] \arrow[u, shift right, "\delta"'] & S_{21} \arrow[l, shift left, "2\,\beta"] \arrow[u, shift right, "\delta"'] \end{tikzcd}$$

After a change of variable in the associated system of differential equations, we obtain

$$s_{00} = (1-m)^3(1-h),\ s_{10} = 3m(1-m)^2(1-h),$$
$$s_{20} = 3m^2(1-m)(1-h),\ s_{30} = m^3(1-h)$$
$$s_{01} = (1-m)^3 h,\ s_{11} = 3m(1-m)^2 h,\ s_{21} = 3m^2(1-m)h,\ s_{31} = m^3 h$$

$$\frac{dm}{dt} = \alpha(1-m) - \beta m, \qquad \frac{dh}{dt} = \gamma(1-h) - \delta h. \tag{3.5}$$

3.3 Cardiac action potential models

The first mathematical model that describes accurately the action potential waveform and the associated permeability changes was proposed by Hodgkin and Huxley in [15]. Although the Hodgkin-Huxley (HH) model was developed specifically for nerve action potentials, the ideas and the mathematical formalism have been used later in a number of models describing cardiac action potentials, see the survey papers [21, 25] and also [9].

3.3.1 Hodgkin-Huxley model

The celebrated Hodgkin-Huxley (HH) model consists of the following system, coupling the equation of the equivalent circuit model (3.1) with the equations of channel gating as in (3.2) - (3.5) for three gating variables m, h, n

$$\begin{cases} C_m \dfrac{dv}{dt} + I_{ion}(v,m,h,n) = I_{app} \\ \dfrac{dm}{dt} = \dfrac{m_\infty(v) - m}{\tau_m(v)} \\ \dfrac{dh}{dt} = \dfrac{h_\infty(v) - h}{\tau_h(v)} \\ \dfrac{dn}{dt} = \dfrac{n_\infty(v) - n}{\tau_n(v)} \end{cases}$$

where the ionic current is the sum of sodium, potassium and leakage currents

$$I_{ion} = I_{Na} + I_K + I_L.$$

Each of these currents has a linear current-voltage structure as in (2.10), with 3 equal activation and 1 inactivation independent subunits for the sodium channel

$$I_{Na} = \bar{G}_{Na} m^3 \; h \; (v - v_{Na}),$$

4 equal activation independent subunits for the potassium channel

$$I_K = \bar{G}_K \; n^4 \; (v - v_K),$$

and no units for the leakage current

$$I_L \quad = \bar{G}_L \; (v - v_L).$$

Here $\bar{G}_{Na}, \bar{G}_K, \bar{G}_L$ are the maximal conductances and v_{Na}, v_K, v_L are the Nernst potentials of each channel type, respectively.

We recall that the coefficients of the gating variables equations are given by

$$m_\infty = \frac{\alpha_m}{\alpha_m + \beta_m}, \qquad \tau_m = \frac{1}{\alpha_m + \beta_m},$$
$$h_\infty = \frac{\alpha_h}{\alpha_h + \beta_h}, \qquad \tau_h = \frac{1}{\alpha_h + \beta_h},$$
$$n_\infty = \frac{\alpha_n}{\alpha_n + \beta_n}, \qquad \tau_n = \frac{1}{\alpha_n + \beta_n},$$

with

$$\alpha_m = \frac{0.1(25-v)}{\exp[0.1(25-v)]-1}, \quad \beta_m = 4\exp\left(\frac{-v}{18}\right),$$
$$\alpha_h = 0.07\exp\left(\frac{-v}{20}\right), \qquad \beta_h = \frac{1}{\exp[0.1(30-v)]+1},$$
$$\alpha_n = \frac{0.01(10-v)}{\exp[0.1(10-v)]-1}, \quad \beta_n = 0.125\exp\left(\frac{-v}{80}\right).$$

The remaining parameters are

$$\bar{G}_{Na} = 120,\ \bar{G}_K = 36,\ \bar{G}_L = 0.3,\ v_{Na} = 115,\ v_K = -12,\ v_L = 10.6.$$

3.3.2 General structure of cardiac cellular membrane models

The ionic current through channels of the membrane is modulated by the transmembrane potential $v = u_i - u_e$, by gating variables $w := (w_1, \ldots, w_M)$ and by ionic intracellular concentration variables $c := (c_1, \ldots, c_S)$. In the membrane models considered in the remainder of this chapter, the ionic current has the following general structure

$$I_{ion}(v,w,c) = \sum_{k=1}^{N} G_k(v,c) \prod_{j=1}^{M} w_j^{p_{j_k}} (v - v_k(c)) + I_n(v,w,c),$$

where N is the number of ionic currents, G_k is the membrane conductance and v_k is the reversal potential for the k-th current, p_{j_k} are integers and I_n accounts for time independent ionic fluxes.

The dynamics of the gating variables w is described in the Hodgkin-Huxley formalism by a system of ODEs having the following structure

$$\begin{cases} \dfrac{dw_j}{dt} = R_j(v,w) = \alpha_j(v)(1-w_j) - \beta_j(v)w_j \\ w_j(0) = w_{j,0} \\ \alpha_j,\ \ \beta_j > 0,\ \ 0 \le w_j \le 1,\ \ j = 1,\ldots,M. \end{cases} \tag{3.6}$$

The dynamics of the ionic concentration variables c is described by the additional system of ODEs

$$\begin{cases} \dfrac{dc_j}{dt} = S_j(v,w,c) = -\dfrac{I_{c_j}(v,w)\cdot A_{cap}}{V_{c_j}\cdot z_{c_j}\cdot F} \\ c_j(0) = c_{j,0} \\ j = 1,\dots,S, \end{cases} \tag{3.7}$$

where I_{c_j} is the sum of ionic currents carrying ion c_j, A_{cap} is the capacitive membrane area, V_{c_j} is the volume of the compartment where c_j is updated, z_{c_j} is the valence of ion c_j and F is the Faraday constant.

The transmembrane current I_m is the sum of the capacitive current, associated with the membrane lipidic bilayer and of the ionic current I_{ion}. Since I_m must balance the applied current I_{app} (see e.g. [16]), then the evolution of the transmembrane potential of a single myocyte is given by the following system of ODEs

$$\begin{cases} C_m\dfrac{dv}{dt} + I_{ion}(v,w,c) = I_m = I_{app}, \\ \dfrac{dw}{dt} - R(v,w) = 0, \qquad \dfrac{dc}{dt} - S(v,w,c) = 0, \\ v(0) = v_0, \qquad w(0) = w_0, \qquad c(0) = c_0, \end{cases} \tag{3.8}$$

where C_m, I_{ion}, and I_{app} are the surface capacitance, the ionic current of the membrane and the applied current per unit area of the membrane surface, respectively.

3.3.3 Reduced ionic models

3.3.3.1 FitzHugh-Nagumo model and other cubic-like models for the excitation phase

In order to investigate phenomena on larger spatial and temporal scales, and also for theoretical purposes, several reduced membrane models have been developed that do not seek to model sub-cellular processes, but only to provide an action potential at a minimal computational cost. The unknowns of these models are a normalized transmembrane potential v and a gating variable w, following the general kinetics

$$\begin{cases} \dfrac{dv}{dt} = f(v,w) \\ \dfrac{dw}{dt} = g(v,w). \end{cases}$$

- FitzHugh-Nagumo model [14]:

$$f(v,w) = -kv(v-a)(v-1) - w, \quad g(v,w) = \varepsilon(v-\gamma w).$$

- Roger-McCulloch model [23]:

$$f(v,w) = -kv(v-a)(v-1) - vw, \quad g(v,w) = \varepsilon(v-\gamma w).$$

- Aliev-Panfilov model [1]:

$$f(v,w) = -kv(v-a)(v-1) - w, \quad g(v,w) = \varepsilon(\gamma v(v-1-a) + w).$$

- Mitchell-Schaeffer model [19]:

$$f(v,w) = -\frac{w}{\tau_{in}} v^2 (v-1) - \frac{v}{\tau_{out}}, \quad g(v,w) = \begin{cases} \dfrac{1-w}{\tau_{open}} & \text{if } v \le v_{gate}, \\ \dfrac{-w}{\tau_{close}} & \text{if } v > v_{gate}. \end{cases}$$

  Here $0 < a < 1,\ k,\ \varepsilon,\ \gamma,\ \tau_{in},\ \tau_{out},\ \tau_{open},\ \tau_{close}, 0 < v_{gate} < 1$ are given constants.

- Morris-Lecar model [20]:

$$\begin{aligned} -f(v,w) &= \frac{g_{Ca} m_\infty (v - v_{Ca}) + g_K w (v - v_K) + g_L (v - v_L)}{C_m} \\ g(v,w) &= \phi\ \frac{w_\infty - w}{\tau_w} \end{aligned} \tag{3.9}$$

  where

$$\begin{aligned} m_\infty(v) &= 0.5[1 + \tanh((v - v_1)/v_2)] \\ w_\infty(v) &= 0.5[1 + \tanh((v - v_3)/v_4)] \\ \tau_w(v) &= 1/\cosh((v - v_3)/(2v_4)). \end{aligned}$$

The parameters of the Morris-Lecar model are: $C_m = 20\mu F/cm^2$, $v_{Ca} = 120\ mV$, $v_K = -84\ mV$, $g_K = 8\ mS/cm^2$, $v_L = -60\ mV$, $g_L = 2\ mS/cm^2$, $v_1 = -1.2\ mV$, $v_2 = 18\ mV$. For the remaining parameters, we consider three different calibrations denoted by **ML1**, **ML2**, **ML3** in Table 3.1 and used later in the section on bifurcation diagrams.

	g_{Ca} (mS/cm^2)	v_3 (mV)	v_4 (mV)	ϕ (ms^{-1})
ML1	4.4	2.	30.	0.04
ML2	4	12	17.4	0.066
ML3	4	12	17.4	0.23

Table 3.1: Three calibrations of the Morris-Lecar parameters g_{Ca}, v_3, v_4, ϕ.

3.3.4 Phase-plane analysis of the FitzHugh-Nagumo (FHN) model

The FitzHugh-Nagumo model can be derived from a simplified electrical circuit model of the cell membrane The circuit consists of a capacitor (with current i_c) in

parallel with a nonlinear current - voltage device (with current $j = F(u)$) and a serial system with a resistor, an inductance, and a battery (with current i). Let $u = u_i - u_e$ be the transmembrane potential and v_R, v_L, v_0 be the resistor, inductance, and battery potentials, respectively. Given an applied current I_a, Kirchhoff's laws state that

$$\begin{cases} i_c + i + j = I_a, \\ u = v_0 + v_L + v_R. \end{cases}$$

From $i_c = C\frac{du}{dt}$, $v_R = Ri$ (Ohm's law), $v_L = L\frac{di}{dt}$ (Faraday's law), we have

$$\begin{cases} C\dfrac{du}{dt} = -i - F(u) + I_a, \\ L\dfrac{di}{dt} = u - Ri - v_0. \end{cases}$$

In order to write this system in dimensionless form, we scale the time as $\tau = R_m t/L$, where $R_m = 1/F'(0)$ and we define the dimensionless variables $v(\tau) = u(L\tau/R_m)/v_m$, $w(\tau) = R_m i(L\tau/R_m)/v_m$, where v_m is the maximal value of u. Then

$$\begin{cases} \dfrac{dv}{d\tau} = \dfrac{L}{v_m R_m}\dfrac{du}{dt} = \dfrac{L}{v_m R_m C}(-i - F(u) + I_a) = \dfrac{L}{R_m^2 C}(-\dfrac{R_m i}{v_m} - \dfrac{R_m F(u)}{v_m} + \dfrac{R_m}{v_m} I_a), \\ \dfrac{dw}{d\tau} = \dfrac{R_m L}{v_m R_m}\dfrac{di}{dt} = \dfrac{L}{v_m L}(v - Ri - v_0) = \dfrac{u}{v_m} - \dfrac{R}{R_m}\dfrac{R_m i}{v_m} - \dfrac{v_0}{v_m}. \end{cases}$$

Defining $\hat{v}_0 = v_0/v_m$, $\hat{F}(v) = \frac{R_m}{v_m}F(v_m v)$, $\hat{I}_a = \frac{R_m}{v_m}I_a$, $\gamma = R/R_m$, $\varepsilon = CR_m^2/L$, we have

$$\begin{cases} \varepsilon\dfrac{dv}{d\tau} = -w - \hat{F}(v) + \hat{I}_a, \\ \dfrac{dw}{d\tau} = v - \gamma w - \hat{v}_0. \end{cases}$$

This system is a particular instance of the generalized FitzHugh-Nagumo planar dynamical system

$$\begin{cases} \varepsilon\dfrac{dv}{d\tau} = f(v,w) + \hat{I}_a, \\ \dfrac{dw}{d\tau} = g(v,w), \end{cases} \tag{3.10}$$

We assume a situation as the one in Fig. 3.3, where the nullcline $f(v,w) = 0$ is a cubic-like function in v with 3 distinct zeros, local maximum value w_{max}, local minimum value w_{min} and the nullcline $g(v,w) = 0$ has only one intersection point (v^*, w^*) with $f = 0$ (for an example with more than one intersection, see Fig. 3.6).

More precisely, we assume that it exists an interval (w_{min}, w_{max}) such that:
- for $w > w_{max}$ $\exists! v : f(v,w) = 0$, defining a function $v = h_-(w)$;
- for $w < w_{min}$ $\exists! v : f(v,w) = 0$, defining a function $v = h_+(w)$;
- for $w_{min} < w < w_{max}$ there are three real roots $h_-(w) < h_0(w) < h_+(w)$ of $f(v,w) = 0$.

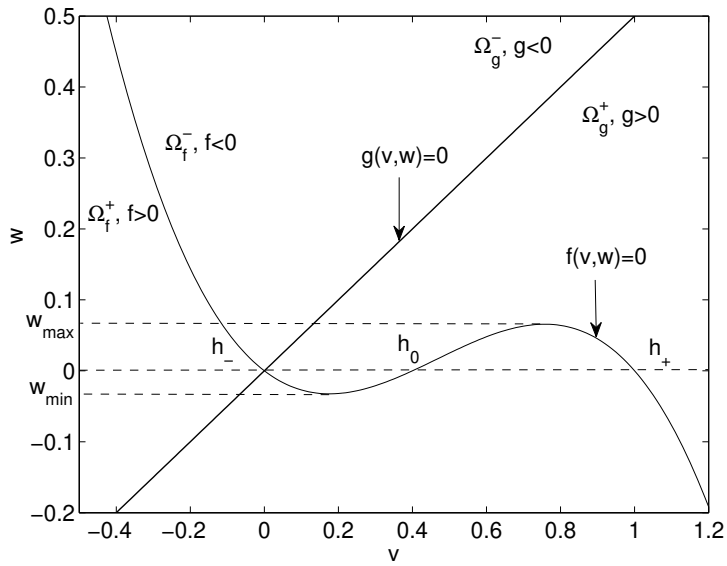


Fig. 3.3: v,w-phase plane, nullclines and notations for the FHN model.

The cubic-like nullcline $f(v,w)=0$ partition the v,w-plane into two regions, $\Omega_f^- = \{(v,w) : w > \overline{w},\ \text{where } f(v,\overline{w})=0\}$ and $\Omega_f^+ = \{(v,w) : w < \overline{w},\ \text{where } f(v,\overline{w}) = 0\}$. We assume that $f<0$ in Ω_f^- and $f>0$ in Ω_f^+. Analogously, the other nullcline $g(v,w)=0$ partitions the plane into two regions $\Omega_g^- = \{(v,w) : w > \overline{w},\ \text{where } f(v,\overline{w})=0\}$ and $\Omega_g^+ = \{(v,w) : w < \overline{w},\ \text{where } f(v,\overline{w})=0\}$, and we assume that $g<0$ in Ω_g^- and $g>0$ in Ω_g^+. These assumptions guarantee that the partial derivatives of f and g are nonpositive or nonnegative but we will make the stronger assumptions $\partial_w f(v,w)<0,\ \partial_v f(v,w)>0$, and $\partial_v g(v,w)>0,\ \partial_w g(v,w)<0$.

According to the chosen initial conditions (v_0,w_0) and the applied current $\hat{I}_a$, the generalized FitzHugh-Nagumo system (3.10) admits solutions with quite different dynamics. The intersection point (v^*,w^*) between the nullclines $f=0$, $g=0$ is an equilibrium point and its stability is determined by the eigenvalues $\lambda_1,\ \lambda_2$ of the Jacobian matrix evaluated at (v^*,w^*)

$$J = \begin{bmatrix} \partial_v f & \partial_w f \\ \partial_v g & \partial_w g \end{bmatrix} :$$

- we have stability if $\lambda_1,\ \lambda_2$ are both real and negative (stable node or sink) or if they are a complex conjugate pair with negative real part (stable spiral);

- we have instability if $\lambda_1,\ \lambda_2$ are both real positive (unstable node or source) or they have opposite signs (saddle) or they are a complex conjugate pair with positive real part (unstable spiral).

Equivalently, the stability of a planar dynamical system is often described in terms of the trace $tr(J) = \lambda_1 + \lambda_2$ and determinant $det(J) = \lambda_1 \lambda_2$ of the Jacobian matrix.

For our generalized FitzHugh-Nagumo system (3.10), we have stability if (v^*, w^*) lies on the lower branch $h_-(w)$ or upper branch $h_+(w)$ of the nullcline $f = 0$, while we have instability if (v^*, w^*) lies on its middle branch $h_0(w)$. In Fig. 3.4, we consider first some examples without applied current, $\hat{I}_a = 0$. We will then vary $\hat{I}_a$ in Fig. 3.5 and in Fig. 3.6 we will consider a case with three equilibrium points. In each figure, the left panel shows the nullclines (dashed lines) and the trajectory (continuous line) in the phase plane v, w starting from the initial point (v_0, w_0) (marked with $\diamond$); the equilibrium point (v^*, w^*) is marked with $\star$.

a) FHN dynamics with zero applied current $\hat{I}_a$ (Fig. 3.4).

a1) Stable equilibrium: subthreshold response. Consider first $w_0 \in (w_{min}, w_{max})$ and $v_0 < h_0(w_0)$. Then v returns to the equilibrium point v^* (Fig. 3.4 top, stable spiral), because in the region where $f < 0$ if the initial point (v_0, w_0) is above the nullcline $g = 0$, then here $g < 0$ and (v, w) returns to the equilibrium point (v^*, w^*), and if (v_0, w_0) is below the nullcline $g = 0$, then here $g > 0$ so w initially increases until the trajectory crosses $g = 0$ into the region where $g < 0$ and (v, w) again returns to (v^*, w^*). Analogous considerations apply when $w_0 \in (w_{min}, w_{max})$ and $v_0 < h_0(w_0)$ in the region where $f > 0$.

a2) Stable equilibrium: action potential (Fig. 3.4 middle, stable spiral). If instead $v_0 > h_0(w_0)$ (for simplicity we still assume $w_0 \in (w_{min}, w_{max})$), then we have an action potential. First, we have an excitation phase during a time $\tau = O(\varepsilon)$, where v tends to the upper branch $h_+(w)$ of the nullcline $f = 0$, because in this region $f > 0$. Indeed, if we scale time as $\tau' = \tau/\varepsilon$ and define $\hat{v}(\tau') = v(\varepsilon\tau')$, $\hat{w}(\tau') = w(\varepsilon\tau')$, we have $\partial_{\tau'}\hat{v} = \varepsilon\partial_\tau v$, $\partial_{\tau'}\hat{w} = \varepsilon\partial_\tau w$, and

$$\begin{cases} \dfrac{d\hat{v}}{d\tau'} = f(\hat{v}, \hat{w}), \\ \dfrac{d\hat{w}}{d\tau'} = \varepsilon g(\hat{v}, \hat{w}). \end{cases}$$

Therefore, for small ε, $\hat{w} \approx w_0$ is approximately constant and $\hat{v}$ evolves toward $h_+(w_0)$. Then for $\tau > O(\varepsilon)$ we have an excited state where (v, w) tends to stay on the upper branch, where $f = 0$ and w evolves approximately according to $\dfrac{dw}{d\tau} = g(h_+(w), w)$. This "plateau" lasts approximately up to the finite time

$$\int_{w_{min}}^{w_{max}} \frac{dw}{g(h_+(w), w)},$$

when w reaches w_{max} at the local maximum of $f = 0$. After this time, we have a recovery phase, where w is again approximately constant and v now evolves toward the lower branch $h_-(w)$ since we are again in the region where $f < 0$. Once the lower branch is reached, (v, w) tends to return to the equilibrium point (v^*, w^*) since the latter lies on the lower branch, i.e. $g(h_-(w^*), w^*) = 0$.

a3) Unstable equilibrium: limit cycle associated with a periodic solution (Fig. 3.4 bottom, unstable spiral). If otherwise (v^*, w^*) lies on the middle branch $h_0(w)$, the equilibrium is unstable and the trajectory of (v, w) starts another excitation phase generating another action potential and so on. In this case, we have a limit cycle associated with a periodic solution.

b) Effect of increasing the applied current $\hat{I}_a$ (Fig. 3.5).

b1) Stable equilibrium: subthreshold response (Fig. 3.5 top, stable spiral). For $\hat{I}_a = 0$, we have a subthreshold response as in case a1) shown in Fig. 3.4, top.

b2) Unstable equilibrium: periodic limit cycle (Fig. 3.5 middle, source). For $\hat{I}_a = 0.15$, the nullcline $f = 0$ is shifted upward and (v^*, w^*) becomes a source point, yielding a periodic limit cycle corresponding to a periodic train of action potentials.

b3) Stable equilibrium (Fig. 3.5 bottom, sink). For $\hat{I}_a = 0.3$, the nullcline $f = 0$ is further shifted upward and (v^*, w^*) becomes a sink point quickly attracting the trajectory to the equilibrium.

c) Multiple equilibrium points (Fig. 3.6). A FHN system with $f(v,w) = v(v+0.1)(1-v) - w,\ g(v,w) = v - 5w,\ \hat{I}_a = 0$, having three equilibrium points is shown in Fig. 3.6: from left to right, they are an unstable spiral, a saddle point, and a sink. The trajectory quickly tends to the lower branch of the $f = 0$ nullcline, follows it until it is repelled by the two unstable equilibrium points, tends to the upper branch of the $f = 0$ nullcline and terminates in the sink.

d) Cathodal and anodal current pulses (Figs. 3.7, 3.8).

In the previous simulations, the action potentials were generated by appropriate choices of initial conditions v_0, w_0 for the transmembrane potential and recovery variables. A more physiological technique to generate an action potential consists in starting from resting initial conditions v_0, w_0 and applying an appropriate current pulse $\hat{I}_a$ for a sufficiently long time interval T_{stim}. In the FitzHugh-Nagumo model (3.10), the application of such current pulse shifts the $f = 0$ nullcline upward if $\hat{I}_a > 0$ (see Fig. 3.7) or downward if $\hat{I}_a < 0$ (see Fig. 3.8). This shift changes temporarily the equilibrium point for the duration of the stimulation interval T_{stim} so that v_0, w_0 is no longer an equilibrium point and the variables v, w start a trajectory toward the closest nullcline branch. When the pulse is turned off at time T_{stim}, the f nullcline returns to the original curve $f = 0$. By this time, v and w have reached a point in phase-space that, for proper choices of pulse amplitude and interval duration, belongs to the excitable region, so that an action potential is generated. Fig. 3.7 shows such an action potential generated by a pulse with $I_a = 0.2$, $T_{stim} = 0.02$, known as cathode make mechanism. Instead Fig. 3.8 shows two cases with action potentials generated by pulses with $I_a = -0.2$, $T_{stim} = 0.1$ (top) or $I_a = -0.2$, $T_{stim} = 0.6$ (bottom), known as anode break mechanism. Additional activation mechanisms (known as cathode break and anode make) are present in space-time cardiac domains, in particular in three-dimensional domains, where more complex virtual electrode polarization (VEP) activation patterns are present, see [30].

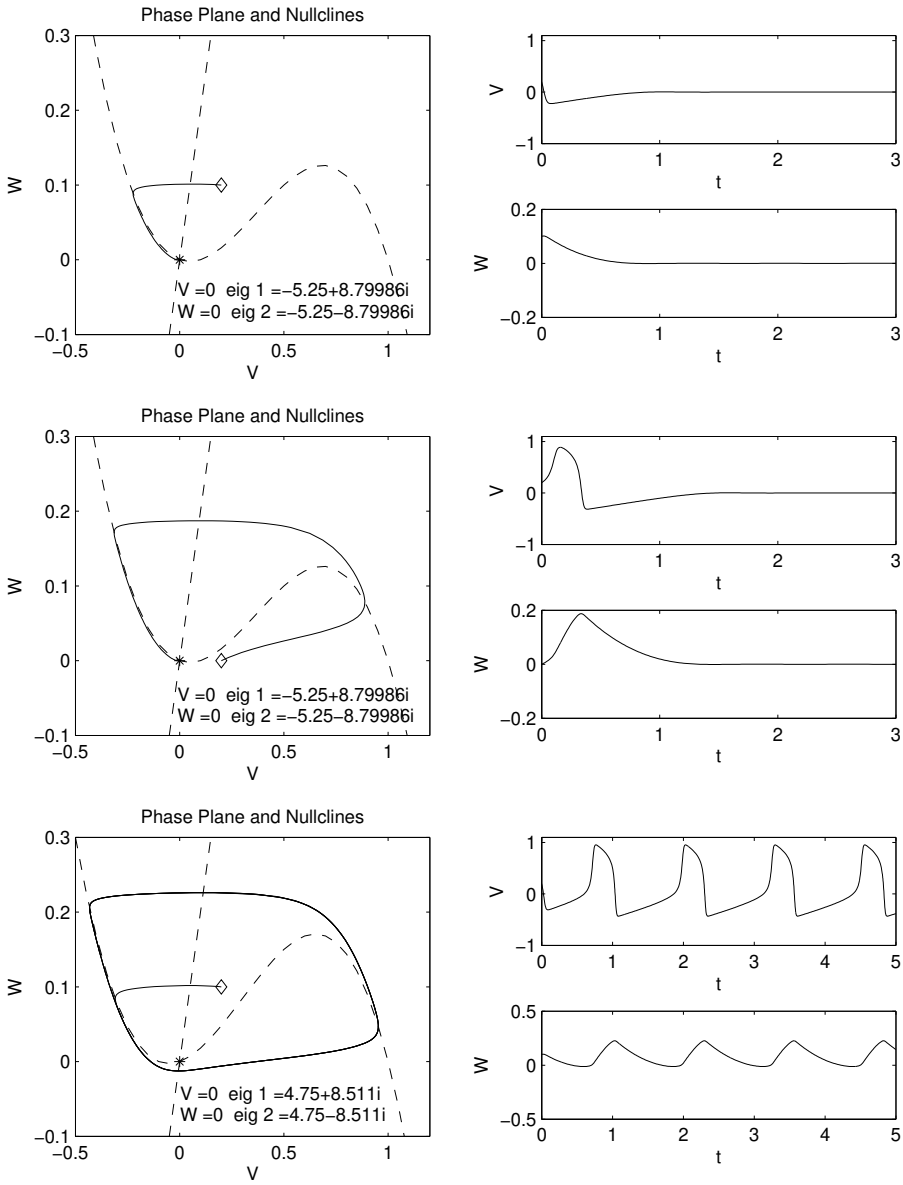


Fig. 3.4: FHN model (3.10) with $\hat{I}_a = 0,\ \varepsilon = 0.01$. In top and middle panels $f(v,w) = v(v-0.1)(1-v) - w,\ g(v,w) = v - w/2$, while in bottom panels $f(v,w) = v(v+0.1)(1-v) - w,\ g(v,w) = v - w/2$. Phase plane trajectories, nullclines, coordinates of the marked equilibrium point $\star$ and associated eigenvalues (left), solutions v, w as functions of time (right).

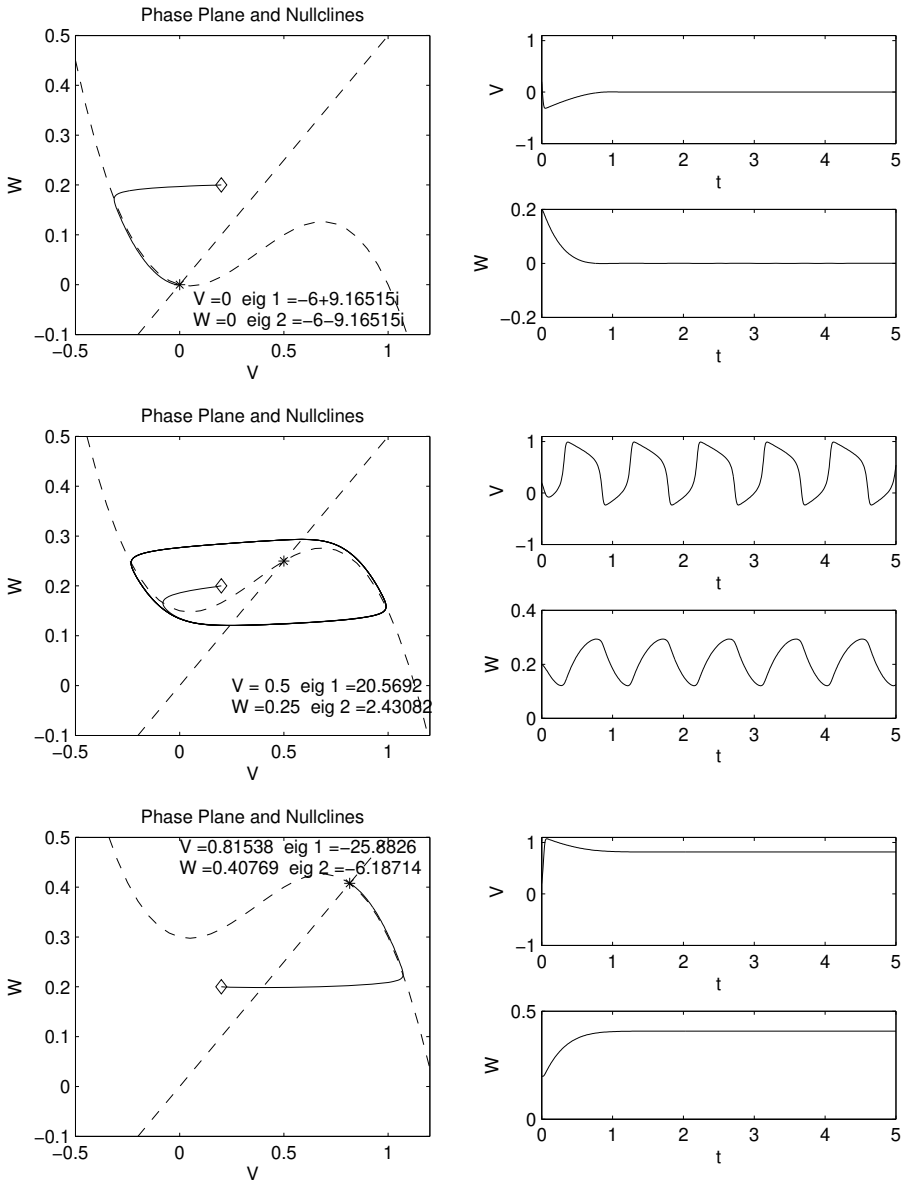


Fig. 3.5: FHN model (3.10) with varying $\hat{I}_a$, with $f(v,w) = v(v-0.1)(1-v) - w$, $g(v,w) = v - 2w$, $\varepsilon = 0.01$. $\hat{I}_a = 0$ (top), 0.15 (middle), 0.3 (bottom). Phase plane trajectories, nullclines, coordinates of the marked equilibrium point $\star$ and associated eigenvalues (left), solutions v, w as functions of time (right).

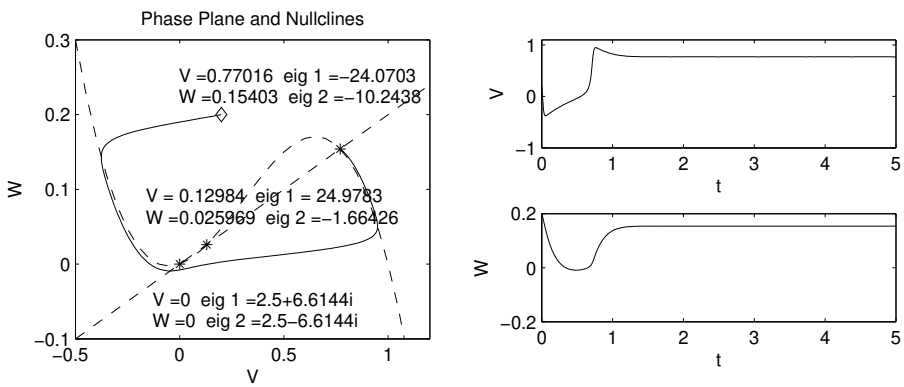


Fig. 3.6: FHN model (3.10) with nullclines intersecting in three points: $f(v,w) = v(v+0.1)(1-v)-w,\ g(v,w) = v-5w,\ \varepsilon = 0.01,\ \hat{I}_a = 0.$. Phase plane trajectories, nullclines, coordinates of the marked equilibrium points $\star$ and associated eigenvalues (left), solutions v,w as functions of time (right).

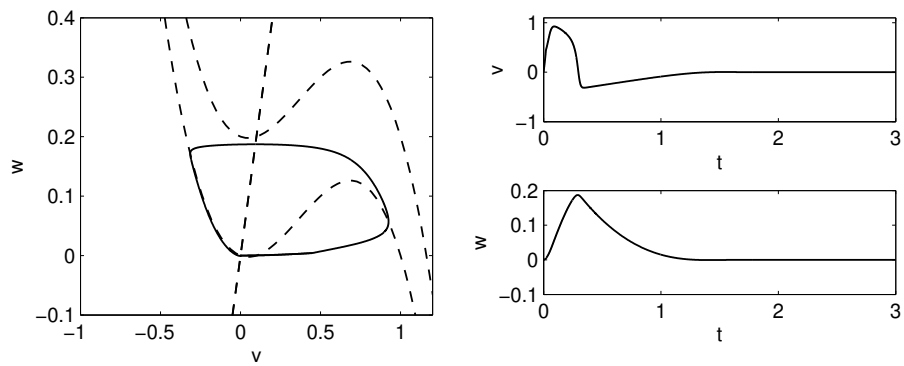


Fig. 3.7: Cathode make due to current pulse $I_a = 0.2$ lasting 0.02 ms for FHN model (3.10) with $f(v,w) = v(v-0.1)(1-v)-w,\ g(v,w) = v-w/2,\ \varepsilon = 0.01$. Phase plane trajectories and nullclines (left), solutions v,w as functions of time (right).

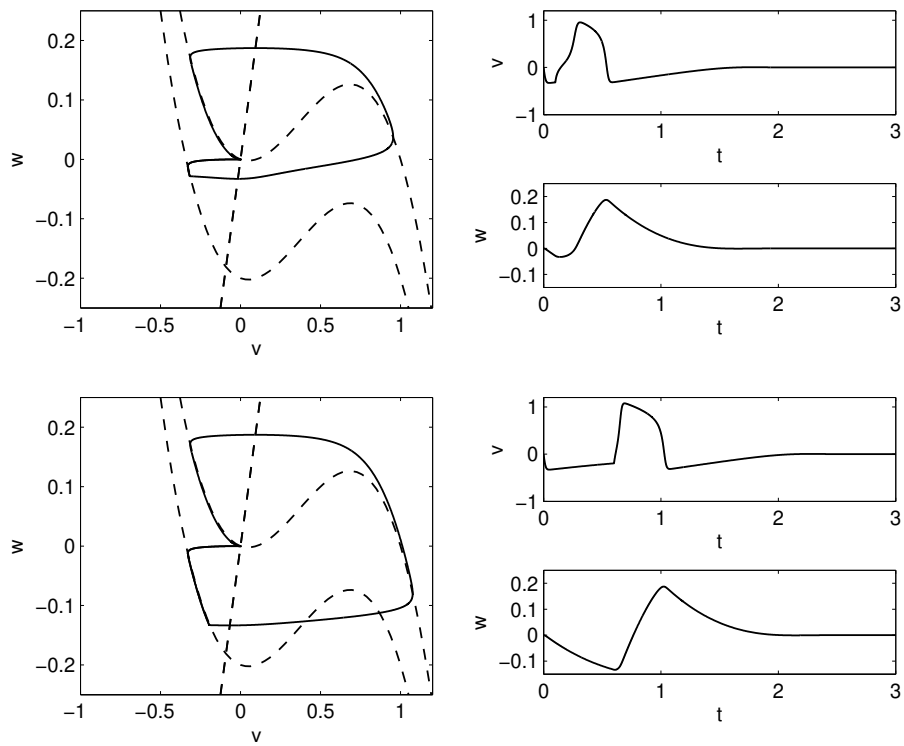
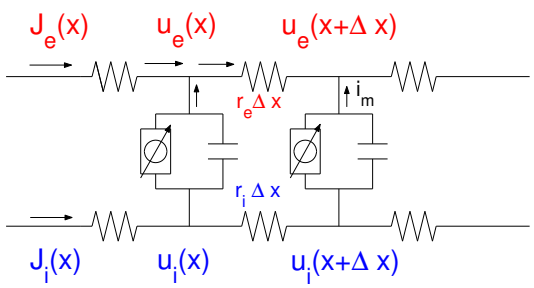


Fig. 3.8: Anode break due to current pulse $I_a = -0.2$ lasting 0.1 ms (top) or 0.6 ms (bottom) for FHN model (3.10) with $f(v,w) = v(v-0.1)(1-v) - w,\ g(v,w) = v - w/2,\ \varepsilon = 0.01$. Phase plane trajectories and nullclines (left), solutions v,w as functions of time (right).

Chapter 4
Models of cardiac fibers

4.1 Cable equation

We model an excitable cell as a long cylindrical cable and assume the so-called core conduction assumption, stating that the potential along the cable depends only on the length variable and not on radial or angular variables, so that the cable model is one-dimensional. Then we assume that the cable is composed of a number of short sections of length dx with isopotential membrane, and that in each section we have only the transmembrane and axial currents, as illustrated in the following diagram



The relationship between the intra- and extracellular current fluxes $J_{i,e}$ ($A\ cm^{-2}$) and the intra- and extracellular potentials $u_{i,e}$ (V) is given by *Ohm's law*

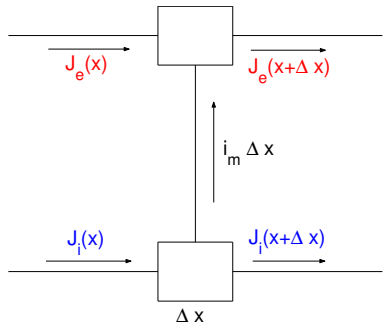
$$J_i = -\frac{1}{r_i}\partial_x u_i = -\sigma_i \partial_x u_i,$$
$$J_e = -\frac{1}{r_e}\partial_x u_e = -\sigma_e \partial_x u_e,$$

where $\sigma_{i,e}$ are the intra- and extracellular conductivities ($\Omega^{-1} cm^{-1}$) and $r_{i,e}$ are the intra- and extracellular resistances per unit space (Ωcm).

We denote by i_m the current across the membrane; it is an outward current, i.e. it is considered positive when directed toward the extracellular space. By *Kirchhoff's law* of conservation of electric charges, we have

$$J_i(x) = J_i(x+\Delta x) + i_m \Delta x,$$
$$J_e(x+\Delta x) = J_e(x) + i_m \Delta x,$$

as illustrated in the following diagram



Assuming J_i, J_e differentiable, in the limit $\Delta x \to 0$ we obtain the following current conservation laws

$$\partial_x J_i = -i_m,$$
$$\partial_x J_e = i_m,$$

which imply

$$\partial_x (J_i + J_e) = 0, \quad \text{i.e.} \quad J_i + J_e = const.$$

We consider a fiber of length L, cut and healed at the two ends, leading to insulating boundary conditions for the intracellular medium

$$J_i = 0 \quad \text{at} \quad x = 0,\ x = L.$$

Moreover, we inject a cathodal extracellular current and eject an anodal extracellular current of strength $I_{app}(t)$ at the two ends of the fiber, i.e.

$$J_e = I_{app}(t) \quad \text{at} \quad x = 0,\ x = L,$$

thus we finally obtain

$$J_i(0) + J_e(0) = J_i(L) + J_e(L) = I_{app}(t), \quad \text{i.e.} \quad J_i(x,t) + J_e(x,t) = I_{app}(t).$$

The membrane current per unit area of the membrane surface is given by

$$I_m = C_m \partial_t v + I_{ion}.$$

This can be expressed as a current i_m per unit volume by multiplying by the ratio χ between the membrane surface area for a given length of fiber and the volume enclosed by this surface, i.e.

$$i_m = \chi\, I_m = \chi\, (C_m \partial_t v + I_{ion}).$$

Collecting together this last equation with Kirchhoff's and Ohm's laws, we have the system

$$\begin{cases} v = u_i - u_e \\ -\sigma_i \partial_x u_i - \sigma_e \partial_x u_e = I_{app}(t) \\ \partial_x(\sigma_i \partial_x u_i) = i_m \\ i_m = \chi(C_m \partial_t v + I_{ion}). \end{cases}$$

By setting $u_e = u_i - v$ in the second equation, we obtain

$$-\sigma_i \partial_x u_i - \sigma_e \partial_x u_i + \sigma_e \partial_x v = I_{app}(t) \;\Rightarrow \partial_x u_i = \frac{1}{\sigma_i + \sigma_e}(\sigma_e \partial_x v - I_{app}(t)),$$

and from the third and fourth equation, we finally obtain the *cable equation*

$$\partial_x(\sigma \partial_x v) - \partial_x \left(\frac{\sigma_i}{\sigma_i + \sigma_e} \right) I_{app}(t) = i_m = \chi(C_m \partial_t v + I_{ion}), \tag{4.1}$$

where

$$\sigma = \frac{\sigma_i \sigma_e}{\sigma_i + \sigma_e}$$

is the conductivity of the bulk medium, corresponding to the conductivity of the intra- and extracellular media connected in series. Due to the presence of gap junctions between two subsequent myocytes, the conductivity $\sigma_i(x)$ is practically constant in the interior of the cell, with value given by the cytosolic conductivity, but in the small length of the gap junctions it undergoes a strong reduction due to the high resistivity of the gap junctions.

4.2 Homogenization

Let us consider an infinite fiber composed by cells connected by end-to-end junctions and embedded in an extracellular equipotential conductor medium, i.e. u_e is constant. Supposing that the cytoplasmic fluid and the junctions are ohmic conductors and applying the current conservation law, in the *stationary regime* the intracellular current density $J_i(x)$ satisfies

$$J_i(x) = -\sigma_i \frac{du_i}{dx}, \qquad \frac{d}{dx} J_i(x) = -i_m(x), \tag{4.2}$$

where i_m is the membrane current. Since the extracellular medium is assumed equipotential, the extracellular flux vanishes and, from $v = u_i - u_e$, we have $\frac{\partial v}{\partial x} = \frac{\partial u_i}{\partial x}$. Setting $\sigma = \sigma_i$ and combining the two equations in (4.2), we get

$$\frac{d}{dx}\left(\sigma \frac{dv}{dx} \right) = i_m. \tag{4.3}$$

The junctions are more resistive than the cytoplasmic fluid, therefore we assume that σ varies periodically with period equal to the distance ε between two subsequent junctions. Hence, $\sigma = \sigma(\frac{x}{\varepsilon})$ is periodic with period ε, or, defining $\xi = \frac{x}{\varepsilon}$, $\sigma = \sigma(\xi)$ is periodic with period 1.

In general, if the cells present some kinds of heterogeneities, we suppose that

$$\sigma = \sigma(x, \xi) \geq \sigma_0 > 0, \tag{4.4}$$

periodic in ξ with period 1.

Combining (4.3) and (4.4), we obtain the following equation, describing the profile of the electric potential along the one-dimensional fiber,

$$\frac{d}{dx}\left(\sigma\left(x, \frac{x}{\varepsilon}\right)\frac{dv}{dx}(x)\right) = i_m(x). \tag{4.5}$$

We have two spatial scales:

- x, varying from 0 to L, length of the fiber, is the macroscopic variable;
- $\xi = \frac{x}{\varepsilon}$, varying in $[0,1]$, describes the singular cellular element and is the adimensional microscopic variable.

Assume now that the solution of equation (4.5) has the following structure

$$v_\varepsilon(x) = V\left(x, \frac{x}{\varepsilon}\right),$$

i.e. it is a function depending on the macro- and microscopic scales, periodic with respect to ξ with period 1.

If $\varepsilon << 1$, we can let $\varepsilon \to 0$ and expand v_ε in power of ε

$$v_\varepsilon(x) = V\left(x, \frac{x}{\varepsilon}\right) = V_0\left(x, \frac{x}{\varepsilon}\right) + \varepsilon V_1\left(x, \frac{x}{\varepsilon}\right) + \varepsilon^2 V_2\left(x, \frac{x}{\varepsilon}\right) + O(\varepsilon^3),$$

with the coefficients $V_i(x, \frac{x}{\varepsilon})$ periodic in ξ of period 1.

We suppose that the coefficients V_i have zero mean value on $\xi \in (0,1)$ for $i \geq 1$,

$$\int_0^1 V_i(x, \xi)\, d\xi = 0 \text{ for } i \geq 1,$$

then it holds

$$v_\varepsilon(x) = \int_0^1 V(x, \xi)\, d\xi = \int_0^1 V_0(x, \xi)\, d\xi.$$

The derivative of v_ε with respect to x is given by

$$\begin{aligned}\frac{d}{dx}v_\varepsilon(x) = \frac{d}{dx}V\left(x, \frac{x}{\varepsilon}\right) &= \frac{\partial}{\partial x}V\left(x, \frac{x}{\varepsilon}\right) + \frac{\partial}{\partial \xi}V\left(x, \frac{x}{\varepsilon}\right)\frac{d\xi}{dx} \\ &= \frac{\partial}{\partial x}V\left(x, \frac{x}{\varepsilon}\right) + \frac{\partial}{\partial \xi}V\left(x, \frac{x}{\varepsilon}\right)\frac{1}{\varepsilon},\end{aligned}$$

thus

$$\frac{d\cdot}{dx} = \frac{\partial\cdot}{\partial x} + \frac{1}{\varepsilon}\frac{\partial\cdot}{\partial\xi}$$

and

$$\begin{aligned}
\frac{d}{dx}\left(\sigma\frac{dv_\varepsilon}{dx}\right) &= \frac{d}{dx}\left(\sigma\left(\frac{\partial V}{\partial x}+\frac{1}{\varepsilon}\frac{\partial V}{\partial\xi}\right)\right)\\
&= \frac{\partial}{\partial x}\left(\sigma\left(\frac{\partial V}{\partial x}+\frac{1}{\varepsilon}\frac{\partial V}{\partial\xi}\right)\right)+\frac{1}{\varepsilon}\frac{\partial}{\partial\xi}\left(\sigma\left(\frac{\partial V}{\partial x}+\frac{1}{\varepsilon}\frac{\partial V}{\partial\xi}\right)\right)\\
&= \frac{\partial}{\partial x}\left(\sigma\frac{\partial V}{\partial x}\right)+\frac{1}{\varepsilon}\frac{\partial}{\partial x}\left(\sigma\frac{\partial V}{\partial\xi}\right)+\frac{1}{\varepsilon}\frac{\partial}{\partial\xi}\left(\sigma\frac{\partial V}{\partial x}\right)+\frac{1}{\varepsilon^2}\frac{\partial}{\partial\xi}\left(\sigma\frac{\partial V}{\partial\xi}\right).
\end{aligned}$$

Imposing that v_ε is solution of (4.5) and expanding in power of ε, we get

$$\begin{aligned}
\frac{d}{dx}\left(\sigma\frac{dv_\varepsilon}{dx}\right) = {} & \frac{1}{\varepsilon^2}\left\{\frac{\partial}{\partial\xi}\left(\sigma\frac{\partial V_0}{\partial\xi}\right)\right\}+\\
& \frac{1}{\varepsilon}\left\{\frac{\partial}{\partial\xi}\left(\sigma\frac{\partial V_1}{\partial\xi}\right)+\frac{\partial}{\partial x}\left(\sigma\frac{\partial V_0}{\partial\xi}\right)+\frac{\partial}{\partial\xi}\left(\sigma\frac{\partial V_0}{\partial x}\right)\right\}+\\
& \left\{\frac{\partial}{\partial\xi}\left(\sigma\frac{\partial V_2}{\partial\xi}\right)+\frac{\partial}{\partial x}\left(\sigma\frac{\partial V_1}{\partial\xi}\right)+\frac{\partial}{\partial\xi}\left(\sigma\frac{\partial V_1}{\partial x}\right)+\frac{\partial}{\partial x}\left(\sigma\frac{\partial V_0}{\partial x}\right)\right\}+\\
& O(\varepsilon^3) = i_m(x).
\end{aligned}$$

If we equal the terms of order $\frac{1}{\varepsilon^2}$, $\frac{1}{\varepsilon}$, 1, we get the following differential problems

$$P0) \qquad \begin{cases} \dfrac{\partial}{\partial\xi}\left(\sigma(x,\xi)\dfrac{\partial}{\partial\xi}V_0(x,\xi)\right) = 0 \ \text{ for } \xi\in(0,1)\\ V_0(x,0) = V_0(x,1), \end{cases} \tag{4.6}$$

$$P1) \qquad \begin{cases} \dfrac{\partial}{\partial\xi}\left(\sigma(x,\xi)\dfrac{\partial}{\partial\xi}V_1(x,\xi)\right) = -\dfrac{\partial}{\partial x}\left(\sigma(x,\xi)\dfrac{\partial}{\partial\xi}V_0(x,\xi)\right)+\\ \qquad\qquad -\dfrac{\partial}{\partial\xi}\left(\sigma(x,\xi)\dfrac{\partial}{\partial x}V_0(x,\xi)\right) \quad \text{for } \xi\in(0,1)\\ V_1(x,0) = V_1(x,1)\\ \displaystyle\int_0^1 V_1(x,\xi)\,d\xi = 0, \end{cases} \tag{4.7}$$

$$P2)\qquad \begin{cases} \dfrac{\partial}{\partial\xi}\left(\sigma(x,\xi)\dfrac{\partial}{\partial\xi}V_2(x,\xi)\right) = i_m(x,\xi) - \dfrac{\partial}{\partial x}\left(\sigma(x,\xi)\dfrac{\partial}{\partial x}V_0(x,\xi)\right) + \\ \qquad\qquad - \dfrac{\partial}{\partial x}\left(\sigma(x,\xi)\dfrac{\partial}{\partial\xi}V_1(x,\xi)\right) + \\ \qquad\qquad - \dfrac{\partial}{\partial\xi}\left(\sigma(x,\xi)\dfrac{\partial}{\partial x}V_1(x,\xi)\right) \qquad \text{for } \xi\in(0,1) \\ V_2(x,0) = V_2(x,1) \\ \displaystyle\int_0^1 V_2(x,\xi)\,d\xi = 0. \end{cases}$$

All the previous problems have the following general structure

$$\begin{cases} \dfrac{\partial}{\partial\xi}\left(\sigma(x,\xi)\dfrac{\partial}{\partial\xi}v(x,\xi)\right) = S(x,\xi) \ \text{ for } \xi\in(0,1) \\ v(x,0) = v(x,1). \end{cases} \tag{4.8}$$

If v is a solution of (4.8), then integrating both the sides of the equation we get

$$\int_0^1 \frac{\partial}{\partial\xi}\left(\sigma\frac{\partial}{\partial\xi}v\right)\,d\xi = \sigma(x,1)\frac{\partial}{\partial\xi}v(x,1) - \sigma(x,0)\frac{\partial}{\partial\xi}v(x,0) = \int_0^1 S(x,\xi)\,d\xi.$$

Since v is periodic, it must hold

$$\sigma(x,1)\frac{\partial}{\partial\xi}v(x,1) - \sigma(x,0)\frac{\partial}{\partial\xi}v(x,0) = 0,$$

from which it follows

$$\int_0^1 S(x,\xi)\,d\xi = 0. \tag{4.9}$$

It can be proved that (4.9) is also a sufficient condition for the solvability of (4.8), i.e.:

if $\int_0^1 S(x,\xi)\,d\xi = 0$, then $\exists!$ solution $v(x,\xi)$ periodic in ξ with period 1 of (4.8).

v is unique up to a constant, that can be fixed imposing zero mean value for $\xi\in(0,1)$,

$$\int_0^1 v(x,\xi)\,d\xi = 0.$$

Problem P0). If we multiply both the sides of (4.6) times V_0 and integrate on $(0,1)$, we get

$$0 = \int_0^1 \frac{\partial}{\partial\xi}\left(\sigma\frac{\partial V_0}{\partial\xi}\right)V_0\,d\xi = -\int_0^1 \sigma\left(\frac{\partial V_0}{\partial\xi}\right)^2\,d\xi$$

and

$$0 = \int_0^1 \sigma\left(\frac{\partial V_0}{\partial\xi}\right)^2\,d\xi \geq \sigma_0\int_0^1\left(\frac{\partial V_0}{\partial\xi}\right)^2\,d\xi,$$

hence

$$\frac{\partial V_0}{\partial \xi} = 0.$$

This means that $V_0 = V_0(x)$ is constant with respect to ξ.

Problem P1). Since $\frac{\partial V_0}{\partial \xi} = 0$, equation (4.7) becomes

$$\frac{\partial}{\partial \xi}\left(\sigma \frac{\partial V_1}{\partial \xi}\right) = -\frac{\partial}{\partial \xi}\left(\sigma \frac{\partial V_0}{\partial x}\right) \Rightarrow \frac{\partial}{\partial \xi}\left(\sigma \frac{\partial V_1}{\partial \xi} + \sigma \frac{\partial V_0}{\partial x}\right) = 0,$$

implying that

$$\sigma\left(\frac{\partial V_1}{\partial \xi} + \frac{\partial V_0}{\partial x}\right) = \phi(x)$$

is constant with respect to ξ. Integrating the resulting equation

$$\frac{\partial V_1}{\partial \xi} + \frac{\partial V_0}{\partial x} = \frac{\phi(x)}{\sigma(x,\xi)} \tag{4.10}$$

over $(0,1)$, we get

$$\frac{\partial}{\partial x} V_0(x) = \phi(x) \frac{1}{\bar{\sigma}(x)},$$

where we have introduced the harmonic average of σ with respect to ξ

$$\bar{\sigma}(x) = \frac{1}{\int_0^1 \frac{1}{\sigma(x,\xi)}\, d\xi}.$$

Equation (4.10) becomes

$$\frac{\partial}{\partial \xi} V_1(x,\xi) = \left(\frac{\bar{\sigma}(x)}{\sigma(x,\xi)} - 1\right) \frac{\partial}{\partial x} V_0(x).$$

Setting

$$V_1(x,\xi) = w(x,\xi) \frac{\partial}{\partial x} V_0(x),$$

it follows

$$\frac{\partial V_1}{\partial \xi} = \frac{\partial w}{\partial \xi} \frac{\partial V_0}{\partial x} = \left(\frac{\bar{\sigma}}{\sigma} - 1\right) \frac{\partial V_0}{\partial x},$$

and then

$$\frac{\partial w}{\partial \xi} = \frac{\bar{\sigma}}{\sigma} - 1.$$

Let us introduce now the cellular problem

$$\begin{cases} \dfrac{\partial}{\partial \xi} w(x,\xi) = \dfrac{\bar{\sigma}(x)}{\sigma(x,\xi)} - 1 \quad \text{for } \xi \in (0,1) \\ w(x,0) = w(x,1) \\ \displaystyle\int_0^1 w(x,\xi)\, d\xi = 0, \end{cases}$$

whose solution is given by

$$w(x,\xi) = w(x,0) + \int_0^\xi \left(\frac{\bar{\sigma}(x)}{\sigma(x,s)} - 1 \right) ds.$$

In order to have

$$\int_0^1 w(x,\xi)\, d\xi = 0,$$

$w(x,0)$ must satisfy

$$\begin{aligned} w(x,0) &= -\int_0^1 \left(\bar{\sigma}(x) \int_0^\xi \frac{1}{\sigma(x,s)}\, ds - \xi \right) d\xi \\ &= -\bar{\sigma}(x) \int_0^1 \int_0^\xi \frac{1}{\sigma(x,s)}\, ds\, d\xi + \frac{1}{2}. \end{aligned}$$

Therefore

$$V_1(x,\xi) = w(x,\xi) \frac{\partial}{\partial x} V_0(x)$$

with

$$w(x,\xi) = \bar{\sigma}(x) \int_0^\xi \frac{1}{\sigma(x,s)}\, ds - \xi - \bar{\sigma}(x) \int_0^1 \int_0^\xi \frac{1}{\sigma(x,s)}\, ds\, d\xi + \frac{1}{2}.$$

Problem P2). To ensure the solvability, it must hold that

$$\begin{aligned} 0 &= \int_0^1 S(x,\xi)\, d\xi \\ &= \int_0^1 \left(i_m(x) - \frac{\partial}{\partial x} \left(\sigma(x,\xi) \frac{\partial}{\partial x} V_0(x) + \sigma(x,\xi) \frac{\partial}{\partial \xi} V_1(x,\xi) \right) - \frac{\partial}{\partial \xi} \left(\sigma(x,\xi) \frac{\partial}{\partial x} V_1(x,\xi) \right) \right) d\xi \\ &= i_m(x) - \frac{\partial}{\partial x} \int_0^1 \sigma(x,\xi) \left(1 + \frac{\partial}{\partial \xi} w(x,\xi) \right) \frac{\partial}{\partial x} V_0(x)\, d\xi - \int_0^1 \frac{\partial}{\partial \xi} \left(\sigma(x,\xi) \frac{\partial}{\partial x} V_1(x,\xi) \right) d\xi. \end{aligned}$$

Since

$$\frac{\partial}{\partial \xi} w(x,\xi) = \frac{\bar{\sigma}(x)}{\sigma(x,\xi)} - 1$$

and the last term

$$\int_0^1 \frac{\partial}{\partial \xi} \left(\sigma(x,\xi) \frac{\partial}{\partial x} V_1(x,\xi) \right) d\xi = \sigma(x,1) \frac{\partial}{\partial x} V_1(x,1) - \sigma(x,0) \frac{\partial}{\partial x} V_1(x,0) = 0$$

because of the periodicity of σ and V_1, we finally get the equation

$$\frac{d}{dx}\left(\bar{\sigma}(x)\frac{d}{dx}V_0(x)\right) = i_m(x).$$

We recall that

$$\int_0^1 V(x,\xi)\,d\xi = \int_0^1 V_0(x,\xi)\,d\xi = \int_0^1 V_0(x)\,d\xi = V_0(x),$$

hence V_0 is denoted as the average potential.

In order to give a detailed approximation of the microscopic scale, one can consider the first order correction

$$V\left(x,\frac{x}{\varepsilon}\right) = V_0(x) + \varepsilon w\left(x,\frac{x}{\varepsilon}\right)\frac{d}{dx}V_0(x) + O(\varepsilon^2).$$

4.3 Traveling waves

A *traveling wave* is a solution of a PDE on an infinite domain that travels at constant velocity with fixed shape. Traveling waves are mainly divided into two types. The first is a *traveling front*, where the wave consists of a transition between two different levels: if v denotes the wave variable, then in front of the wave, v is steady at some low value, and behind the wave, v is steady at a higher value. The second type of traveling wave is a *traveling pulse*, where the wave begins and ends at the same value of v and resembles a moving bump. Thus, the main difference between traveling fronts and traveling pulses is that the former do not present recovery, while in the latter recovery plays an important dynamic role; see e.g. [5, 17].

4.3.1 Traveling fronts

The cable equation (4.1) has been extensively studied in the special version known as the *bistable equation*

$$\partial_t v = \partial_{xx} v + f(v), \tag{4.11}$$

where

- $f(v)$ has three zeros at $v = 0, \alpha, 1$, with $0 < \alpha < 1$;
- $f'(0) < 0$ and $f'(1) < 0$.

The values $v = 0$ and $v = 1$ are stable steady solutions of the ODE $\partial_t v = f(v)$. For existence, uniqueness and stability of traveling wavefront solutions and long time behavior of the solution of (4.11), see [13, 11, 12].

An example of f often used in this context is the cubic polynomial

$$f(v) = av(v-1)(\alpha - v), \qquad 0 < \alpha < 1.$$

A traveling wave is a translation-invariant solution of (4.11), providing a transition between the two stable rest states (zeros of $f(v)$) and it travels with a constant speed c, i.e. a solution of the form

$$v(x,t) = V(x+ct) = V(\xi), \tag{4.12}$$

for a suitable value of c. When written as a function of the new traveling wave variable ξ, the wave appears stationary. Because we use $\xi = x + ct$, a solution with a positive c corresponds to a wave moving from right to left, while a solution with a negative c moves from left to right.

By imposing that $v(x,t)$ given in (4.12) satisfies the bistable equation (4.11), one obtains that any traveling wave solution solves the following second order ODE

$$V_{\xi\xi} - cV_\xi + f(V) = 0. \tag{4.13}$$

If $V(\xi)$ provides a transition between rest values, it must hold that

$$\lim_{\xi \to \pm\infty} f(V(\xi)) = 0.$$

From (4.13), it follows that a traveling wave solution of the bistable equation is a solution of the first order ODE system

$$\begin{aligned} V_\xi &= W \\ W_\xi &= cW - f(V), \end{aligned}$$

connecting the steady states $(V,W) = (0,0)$ and $(V,W) = (1,0)$ in the (V,W) phase plane. Such a trajectory, connecting two different steady states, is called a *heteroclinic trajectory* and it approaches $(0,0)$ as $\xi \to -\infty$ and $(1,0)$ as $\xi \to +\infty$. Beyond the steady states $(0,0)$ and $(1,0)$, which are saddle points, the other steady state is $(\alpha,0)$, which is a node, if c is positive, or a spiral point, if c is negative. Because $(0,0)$ and $(1,0)$ are saddle points, in order to find a traveling wave solution, we determine whether a parameter c exists such that the trajectory leaving $(0,0)$ at $\xi = -\infty$ reaches $(1,0)$ at $\xi = +\infty$.

We first determine the sign of velocity c. Supposing a monotone increasing ($V_\xi > 0$) connecting trajectory exists, multiplying (4.13) by V_ξ and integrating from $\xi = -\infty$ to $\xi = +\infty$, one obtains the following relationship

$$c \int_{-\infty}^{+\infty} W(\xi)^2 d\xi = \int_0^1 f(u)du. \tag{4.14}$$

Thus, if the integral of $f(u)$ between $u = 0$ and $u = 1$ is positive, the traveling waves move from right to left and the state variable V moves from $V = 0$ to $V = 1$, while if the integral is negative viceversa.

Theorem 4.1. *Suppose* $\int_0^1 f(u)du > 0$. *Then there exists a unique velocity* $c > 0$ *with a corresponding traveling wave solution of the bistable equation.*

Proof. To verify uniqueness, from

$$\frac{dW}{dV} = c - \frac{f(V)}{W},$$

we observe that the slope $\frac{dW}{dV}$ of trajectories in the (V,W) phase plane is a monotone increasing function of the parameter c. Let $c = c_0$ a velocity value for which a connecting trajectory exists. Suppose now that for a value $c_1 > c_0$, another connecting trajectory exists. The trajectory leaving the saddle point $(0,0)$ for c_1 must lie above the connecting trajectory for c_0. For the same reason, the trajectory approaching the saddle point $(1,0)$ for c_1 must lie below the connecting trajectory for c_0. A single trajectory cannot lie above and below another one, thus there cannot be a connecting trajectory for $c > c_0$. Analogously, there cannot be a connecting trajectory for $c < c_0$. Hence the connecting trajectory is unique.

To verify existence, we first suppose c large. Let K be the smallest positive number for which $\frac{f(u)}{u} \leq K,\ \forall u:\ 0 < u \leq 1$ and let σ be any fixed positive number. On the line $W = \sigma V$ the slope of trajectories satisfies

$$\frac{dW}{dV} = c - \frac{f(V)}{W} = c - \frac{f(V)}{\sigma V} \geq c - \frac{K}{\sigma}.$$

With c large enough, it is assured that $c - \frac{K}{\sigma} > \sigma$. Therefore once trajectories are above the line $W = \sigma V$, they stay above it. Because the slope of the unstable trajectory leaving the saddle point $(V,W) = (0,0)$ is the positive root of $\lambda^2 - c\lambda + f'(0) = 0$, which is always larger than c, this trajectory starts above the curve $W = \sigma V$. Thus is stays always above $W = \sigma V$ and consequently passes above the saddle point $(V,W) = (1,0)$.

With $c = 0$, by multiplying (4.13) by V_ξ and integrating, one obtains that the trajectory leaving $(V,W) = (0,0)$ satisfies

$$\frac{W^2}{2} + \int_0^V f(u)du = 0,$$

and if it reaches $V = 1$ for some value of W, then

$$\frac{W^2}{2} + \int_0^1 f(u)du = 0,$$

which can occur only if $\int_0^1 f(u)du \leq 0$. Since this is in contradiction with the hypothesis that $\int_0^1 f(u)du > 0$, it follows that the trajectory cannot reach $V = 1$ and neither it can remain in the first quadrant, because $W > 0$ implies that V increases. As a consequence, this trajectory must intersect $W = 0$ for some $V < 1$, thus it cannot be the connecting trajectory.

Summarizing, we have found two trajectories, one with c large, which misses the saddle point $(V,W) = (1,0)$ staying above it, and one with $c = 0$, which intersects the $W = 0$ axis for some $V < 1$. Because trajectories depend continuously on the parameters of the problem, there is a family of trajectories depending continuously on c between these two special trajectories, and therefore there must be at least one trajectory reaching $(V,W) = (1,0)$.

Example. Suppose $f(v)$ is the cubic polynomial

$$f(v) = v(v-\alpha)(1-v).$$

Our goal is to find a heteroclinic connection between 0 and 1, so we guess that

$$W = BV(1-V). \tag{4.15}$$

By substituting (4.15) into (4.13), it follows that it must hold

$$(-2B^2+1)V + (B^2 - cB - \alpha) = 0,$$

which is identically zero only if we choose

$$B = \frac{1}{\sqrt{2}} \qquad c = \frac{1}{\sqrt{2}}(1-2\alpha),$$

showing that the velocity is a decreasing function of α and the direction of propagation changes at $\alpha = 1/2$, according to the change of sign of $\int_0^1 f(u)du$. From the solution of (4.15), we obtain the profile of the traveling wave

$$V(\xi) = \frac{1}{2}\left[1 + \tanh\left(\frac{x}{2\sqrt{2}}\right)\right].$$

4.3.2 Traveling pulses

A *traveling pulse* is a traveling wave solution that starts and ends at the same steady state of the governing equations. In the previous section, we have seen that a traveling front solution of the bistable equation corresponds to a *heteroclinic trajectory* in the (U,W) phase plane, i.e. a trajectory that connects two different steady states of the system. Analogously, a traveling pulse corresponds to a trajectory that starts and ends at the same steady state in the (U,W) phase plane. Such trajectories are called *homoclinic orbits*.

In order to study wave propagation in excitable media, we first consider the following FitzHugh-Nagumo system

$$\begin{aligned} \varepsilon\partial_t v &= \varepsilon^2\partial_{xx}v + f(v,w) \\ \partial_t w &= g(v,w), \end{aligned} \tag{4.16}$$

where ε is a small positive number and, without any loss of generality, the space variable x has been scaled so that the diffusion coefficient is ε^2, see e.g. [13, 11, 12, 29]. This type of reaction-diffusion singular perturbation problem develops in a short time a sharp propagating interface, i.e. a steep wavefront. In fact, in [7, 8], it was shown that, after a time period of order $O(\varepsilon \log \varepsilon)$, a sharp transition layer of thickness of order $O(\varepsilon)$ appears. Let us recall that the variable v represents the transmembrane potential, while w represents a slow gating variable.

We now proceed further in understanding the structure of traveling wave solutions of (4.16), by performing a singular perturbation analysis, see e.g. [6]. Starting from smooth initial data $(v_0(x), w_0(x))$, the diffusion term $\varepsilon \partial_{xx} v$ and the variation of w from its initial data can be neglected for a short time. In fact, setting $\tau = \frac{t}{\varepsilon}$, $\hat{v}(x,\tau) = v(x,\varepsilon t)$ and $\hat{w}(x,\tau) = w(x,\varepsilon t)$, then

$$\begin{aligned} \partial_t \hat{v} &= \varepsilon^2 \partial_{xx} \hat{v} + f(\hat{v},\hat{w}) \\ \partial_t \hat{w} &= \varepsilon g(\hat{v},\hat{w}). \end{aligned} \tag{4.17}$$

An approximation of order ε is obtained setting ε to zero in (4.17), i.e.

$$\begin{aligned} \partial_t \hat{v} &= f(\hat{v},\hat{w}) \\ \partial_t \hat{w} &= 0. \end{aligned}$$

Thus $\hat{w}(x) = w_0(x)$ and, using the notations introduced in Section 3.3.4 (see also Fig. 3.3), in a short time ($t = O(\varepsilon)$) $\hat{v} \to h_\pm(w_0)$, if $v_0(x) \gtrless h_0(w_0)$. The region of space where $v = h_+(w)$ is called *excited region*, while the region where $v = h_-(w)$ is called *recovered region*. In these two regions, it holds

$$\begin{aligned} f(\hat{v},\hat{w}) &= 0 \\ \partial_t \hat{w} &= g(\hat{v},\hat{w}), \end{aligned} \tag{4.18}$$

thus

$$\begin{aligned} \hat{v} &= h_\pm(\hat{w}) \\ \partial_t \hat{w} &= G_\pm(\hat{w}) = g(h_\pm(\hat{w}),\hat{w}). \end{aligned}$$

However, there are regions of space, called *interfaces*, where diffusion is large and (4.18) cannot be applied. To understand what happens when diffusion is large, we rescale space and time by setting $\tau = t$ and $\xi = \frac{x-y(t)}{\varepsilon}$, where $y(t)$ denotes the position of the wavefront, i.e. the points

$$y(t) : v(y(t),t) = h_0(w_0(x)).$$

The original system of equations (4.16) becomes

$$\begin{aligned} \hat{v}_{\xi\xi} + y'(\tau)\hat{v}_\xi + f(\hat{v},\hat{w}) &= \varepsilon \partial_\tau \hat{v} \\ y'(\tau)\hat{w}_\xi &= \varepsilon(g(\hat{v},\hat{w}) - \partial_\tau \hat{w}), \end{aligned}$$

and, after setting $\varepsilon = 0$, it reduces to

$$\begin{aligned} \hat{v}_{\xi\xi} + y'(\tau)\hat{v}_\xi + f(\hat{v},\hat{w}) &= 0 \\ y'(\tau)\hat{w}_\xi &= 0. \end{aligned} \tag{4.19}$$

It follows from (4.19) that $\hat{w}$ is independent of ξ, but not necessarily of τ, i.e. $\hat{w}(\tau) = \hat{w}(y(\tau),\tau)$. Because (4.19) describes the transition layer between regions where (4.18) holds, we require the *matching condition*

$$\lim_{\xi\to\pm\infty} f(\hat{v}(\xi,\tau), w(\tau)) = 0,$$

i.e.

$$\hat{v}(\pm\infty,\tau) = h_\mp(\hat{w}(\tau)).$$

Since $y(\tau)$ represents the position of the wavefront, $y'(\tau)$ is the wave velocity.

The first equation in (4.19) is a *bistable equation*, in the form (4.13), thus, for fixed $\hat{w}$, there exists $c = c(\hat{w})$ for which the equation

$$\hat{v}_{\xi\xi} + c(\hat{w})\hat{v}_\xi + f(\hat{v},\hat{w}) = 0 \tag{4.20}$$

has a *heteroclinic orbit* connecting two stable roots of $f(\hat{v},\hat{w}) = 0$. This means that $c(\hat{w})$ is the unique value for which (4.20) has a solution with $\hat{v} \to h_+(\hat{w})$ as $\xi \to -\infty$ and $\hat{v} \to h_-(\hat{w})$ as $\xi \to +\infty$.

Summarizing, in most of space, (4.18) holds. At any transition between the two types of (4.18), a sharp transition in v occurs, with a wavefront traveling at the speed $y'(t) = c(w)$ if $v = h_+(w)$ on the left and $v = h_-(w)$ on the right, or $y'(t) = -c(w)$ if $v = h_-(w)$ on the left and $v = h_+(w)$ on the right. We now suppose that far to the right the medium is at rest, i.e. $\exists\ w_+ :\ G_-(w_+) = 0$, and that a wavefront of excitation moves from left to right with velocity $y'(t) = c(w_+) > 0$. Following the same procedure used to derive (4.14), one obtains

$$c(w) = \frac{\int_{h_-(w)}^{h_+(w)} f(v,w)dv}{\int_{-\infty}^{+\infty} v_\xi^2 d\xi},$$

and thus $c(w_+) > 0$ if and only if

$$\int_{h_-(w)}^{h_+(w)} f(v,w)dv > 0. \tag{4.21}$$

If (4.21) fails to hold, then the medium is not sufficiently excitable to sustain a traveling pulse.

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